

Presentation Script

A Study of Large Language Models' Emotion Classification in Ambiguous Text
with Perceived Authorship

Plain-language walkthrough of the three analysis notebooks, with every graph
explained

Notebook 00: Corpus and sampling (how the 300 texts were chosen)

Notebook 01: From majority labels to emotion distributions (the hypothesis tests)

Notebook 02: What changed inside the distributions (the deeper analysis)

All numbers come from the executed notebooks in *thesis final/distribution_analysis_v1/* and the tables in *thesis final/results/*. Each graph section has three parts: what the graph shows, what it means, and a shaded *Say it like this* line you can read out.

How to use this script

The talk follows the same order as the notebooks:

- **Part 1: What you need to know first.** Every idea and calculation used in the analysis, one step at a time, with tiny worked examples. Read this before anything else.
- **Part 2: The study in one page** (question, design, hypotheses).
- **Part 3: Notebook 00:** how the texts were chosen and grouped.
- **Part 4: Notebook 01:** first the old one-label method, then the vector method, then both compared.
- **Part 5: Notebook 02:** what exactly moved inside the distributions.
- **Part 6: The answers** to every research question and hypothesis, with the reasoning.
- **Part 7: Likely questions** from the audience, with short answers.

All calculations described here are written out in the notebooks themselves, just before the result that uses them. If time is short, present Parts 2 and 6, and show Graphs 5, 8, 12, 14, 16, 17 and 18 from Notebook 01. Those carry the whole argument. Part 1 is for your own preparation.

Part 1: What you need to know first

Read this part before the graphs. Each idea builds on the one before, and each one is exactly what the notebook code does. The examples use tiny made-up votes, so every number can be checked by hand. The last page is a checklist of where each idea is used.

1. The basic units: votes, versions and comparisons

- A **vote** is one emotion chosen by one person or one model.
- Every text was shown in several **versions** (also called conditions): **neutral** (no author), **persona A**, **persona B**, and for models also **generic** ("written by a person").
- A **comparison** is one text, neutral against one persona. Humans: 300 texts × 2 personas = **600** comparisons. Models: × 6 prompt set-ups = **3,600**.
- These are still only **300 different texts**. The 12 model comparisons of one text are not 12 independent pieces of evidence. So every average, error bar and test treats the **text** as the unit: each text is first averaged, then texts are compared.

2. From votes to vote shares (the vector)

Count the votes for each of the 11 emotions and divide by the number of votes. The result is a list of 11 percentages that adds up to 100%. This list is the **vote-share vector** (called a probability distribution in the thesis).

Votes	Vote shares (only non-zero emotions shown)
anger, anger, sadness	anger 67%, sadness 33%
joy, joy, joy	joy 100%
fear, relief, trust	fear 33%, relief 33%, trust 33%

Model votes work the same way: 7 models give up to 7 votes. Answers outside the 11 emotions are left out, so a vector can rest on fewer votes.

3. Majority, plurality, tie, and the old method's single label

Votes (5 readers)	Name	Why
anger ×3, fear, joy	Majority	more than half (3 of 5) chose anger
anger ×2, fear, joy, trust	Plurality	anger is on top but has less than half
anger ×2, fear ×2, joy	Tie	two emotions share the top
five different emotions	No top at all	everyone disagrees

The **old method** keeps only the top emotion, the **selected label**. For a tie it picks the alphabetically first emotion. So *surprise, joy, anger* (a three-way tie) becomes "anger", even though only 1 of 3 people chose it. With the reverse alphabet it would be "surprise". The old method records a **label change** (yes or no) when the selected label differs between neutral and persona.

4. The shift: total variation distance (TV)

The shift measures **how much of the vote share has to move** to turn the neutral votes into the persona votes. Take the difference for every emotion, add them all up (ignoring minus signs), then halve it. Halving is needed because every vote that leaves one emotion arrives at another, so otherwise each move would be counted twice.

Worked example. Neutral: anger, anger, sadness. Persona: anger, fear, fear.

Emotion	Neutral	Persona	Difference
anger	67%	33%	33
sadness	33%	0%	33
fear	0%	67%	67
		Sum	133
		Shift = sum ÷ 2	67%

- 0% means identical votes; 100% means completely different emotions.
- Same answer another way: add up only the gains (fear +67). That also gives 67%.
- In this example the old method also records a label change (anger → fear).

5. Why a single label hides change

Neutral: trust, trust, trust (100% trust). Persona: trust, trust, joy, pride, surprise (40% trust, 20% each for joy, pride and surprise). The winner is **trust on both sides**, so the old method records **no change**. The shift is **60%**. This is a real text from the study (Graph 12). The vector method was introduced to see changes like this.

6. The five kinds of change

Kind	Example (neutral → persona)
No change	joy, joy, pride → joy, joy, pride
Same emotions, new percentages	anger 67%, joy 33% → anger 80%, joy 20%
Emotions appear only	trust 100% → trust 40%, joy 20%, pride 20%, surprise 20%
Emotions disappear only	fear 67%, relief 33% → fear 100%
Appear and disappear	anger, anger, sadness → anger, fear, fear (sadness goes, fear comes)

"Appear" and "disappear" describe the **group's** votes. Different people rated each version, so it does not mean one person changed their mind.

7. The shift vector and its direction (cosine)

The **shift vector** (delta) is persona shares minus neutral shares, emotion by emotion. In the example from section 4: **anger -33, sadness -33, fear +67**. It always adds up to zero, because every vote that is lost somewhere is gained somewhere else.

Cosine compares the *direction* of two shift vectors, ignoring their size: **+1** = they point the same way, **0** = unrelated, **-1** = opposite.

Worked example: same direction, different size. The 7 models answer anger ×5, sadness ×2 without a persona, and anger ×4, sadness ×1, fear ×2 with it.

	anger	sadness	fear	Shift size
Humans (section 4)	−33	−33	+67	67%
Models	−14	−14	+29	29%

Both move away from anger and sadness towards fear in the same proportions, so the cosine is **1.00**. But humans moved 67% and models only 29%. **Direction and size are separate questions**, and that is exactly what H4 asks. If the models had instead moved *towards* anger and sadness and away from fear, the cosine would be −1.

If either side did not move at all, its shift vector is all zeros and has no direction. Those comparisons are **left out** of the direction analysis, never counted as zero. That is why direction is measurable in only 2,020 of the 3,600 comparisons.

8. Spread (entropy)

Entropy measures how spread out the votes are. It is not the same as how far they moved.

Votes	Entropy
joy, joy, joy (everyone agrees)	0
joy, joy, pride	0.64
joy, pride, relief (all different)	1.10

A **positive entropy change** under a persona means the votes became more spread out: the raters of that version agreed less with each other.

9. Why a shift is never zero, even when nothing happens

Different people rated the neutral and persona versions. Three people rarely give identical answers, so the votes would differ even if the persona had **no effect at all**. Unambiguous texts shift by about 45% on average for humans, yet no persona effect is detected for them. So a raw shift cannot be judged on its own. It has to be compared with **what chance alone would produce**.

10. The chance shuffle for humans (worked example)

If the persona did nothing, it would not matter which version each vote came from. So:

- put the neutral and persona votes of one text into a single pool;
- split the pool back into two groups of the original sizes, in every possible way;
- measure the shift for each split.

With the section 4 example the pool is anger ×3, sadness ×1, fear ×2. There are exactly **20** ways to split 6 votes into 3 + 3:

Shift in that split	Number of splits
33%	12
67%	6
100%	2

Shift in that split	Number of splits
Average (what chance gives)	50%

- The average, **50%**, is the **chance expectation** for this text. It is the orange × in the H1/H2 graphs ("conditional-null mean").
- The observed shift was 67%, so the **extra shift** (excess over chance) is **+17 points**.
- 8 of the 20 splits reach 67% or more. On its own, this one text is weak evidence (8 in 20 = 0.40). The tests therefore combine many texts (section 12).

11. The chance shuffle for models

The same model answered both versions, so votes cannot be swapped *between* models. Instead, each model's own two answers are either kept or swapped (like flipping a coin for each model). With 7 models there are $2 \times 2 \times \dots = 128$ possibilities.

Consequence: if only **one** model changed its answer, swapping gives the same shift every time, so that comparison cannot count as evidence. In 61% of model comparisons at most one model changed. The model test (H2) therefore detects changes that **several models share**. That is why the individual models' change rates are reported next to it (Graph 27).

12. From one text to a test: the p-value

- **Observed:** the average shift over the 100 texts of a group (e.g. author-relevant, persona A).
- **Chance:** 10,000 times, pick one random split per text (section 10) and average them. That gives 10,000 averages that could happen if the persona did nothing.
- **p** = the share of those 10,000 chance averages that are at least as big as the observed one.
- **p below 0.05** (1 in 20) is called **significant**: chance alone rarely produces a result this big.

Two cautions. p is **not** the size of an effect; the size is the shift or the excess. And a large p does **not** prove there is no effect: it means the data could not detect one. Say "not detected", never "no effect".

The old method uses exactly the same procedure, but measures "did the label change (1 or 0)" instead of the shift.

13. Many tests at once: Holm and BH corrections

Each test has a 1 in 20 chance of a lucky pass. Run 6 tests and the chance that at least one passes by luck is much higher. Corrections raise the p-values to compensate. **Holm** (used for the hypothesis tests): sort the p-values, multiply the smallest by 6, the next by 5, and so on, and never let an adjusted value be smaller than the one before it.

Worked example with the real H1 results:

Test (H1, vector method)	Raw p	× factor	Holm p	Significant?
Author-relevant, A	0.0011	× 6	0.0066	Yes
Author-independent, A	0.0022	× 5	0.0110	Yes
Author-relevant, B	0.0028	× 4	0.0112	Yes
Unambiguous, A	0.0660	× 3	0.198	No
Author-independent, B	0.0679	× 2	0.136 → 0.198*	No

Test (H1, vector method)	Raw p	× factor	Holm p	Significant?
Unambiguous, B	0.2145	× 1	0.2145	No

* raised to 0.198 because an adjusted value may not be smaller than the one above it. That is why two different tests both show $p = 0.198$.

BH (Benjamini-Hochberg) is a gentler correction for large *exploratory* families, such as the 66 per-emotion tests: the k -th smallest p -value is multiplied by $66 \div k$.

14. Error bars (95% ranges)

An error bar shows the plausible range of an average. It is made by **resampling texts**: draw 100 texts at random from the 100 (the same text can be drawn twice), recompute the average, repeat 10,000 times, and keep the middle 95% of the results. Because whole texts are drawn, a text's repeated comparisons stay together.

If a range for a *difference* excludes zero, the difference is clearly there. Two *separate* ranges that overlap are only a hint, not a test.

15. Comparing two groups of texts (H3)

- Give each text one number: its average shift over the two personas.
- **Difference** = author-relevant average minus author-independent average.
- **p**: shuffle which group each text belongs to 10,000 times and count how often the difference is at least as big (in either direction).
- **Raw version** (the main test): uses the observed shifts.
- **Chance-adjusted version** (a sensitivity check): first subtract each text's own chance expectation (section 10), then compare what is left. Ambiguous texts have raters who disagree more to begin with, so their chance expectation is higher. This check was added after the raw result was known, so it is reported next to the main test, not instead of it.

16. Three more tools, in brief

Tool	What it does	Used for
Sign-flip test	Randomly flip the sign (+/−) of each text's change, 10,000 times; see how often the average is as far from zero as the real one	Per-emotion gains and losses (Graph 25)
Friedman test	Within each text, rank the 6 prompt set-ups; if the set-up did not matter, all would get similar average ranks	Prompt set-ups (Graph 28)
Equal-vote-count check	Randomly keep 3 votes per human version and the same 3 models on both model sides, 200 times, then recompute	Is the human–model gap just about vote counts? (Graph 26)

17. Reading results honestly

Say	Not
"No effect detected"	"No effect"

Say	Not
"Significant for A, not for B; A and B were not tested against each other"	"The persona works for A but not for B"
"Main test" vs "sensitivity check added afterwards"	Treating both as equal, or swapping them
"Is associated with a larger shift"	"Causes" / "explains X%" (for the generic-author control)
"Descriptive" (per-model rates, Graph 27; human–model match, Graph 24)	"Model X is significantly more sensitive" / "humans and models agree"
"Supported at the ensemble level" (H2)	"Every model is affected" as a tested claim

Checklist: where each idea is used

Idea (section)	Where it matters
Comparison, text as unit (1)	Every graph and test
Vote shares (2), shift (4)	Graphs 13–18, H1–H4 with vectors
Selected label, ties (3)	Graphs 5–10 (old method), tie check
Hidden change (5), five kinds (6)	Graphs 11, 12, 15, 19–24
Shift vector and cosine (7)	H4 direction (Graph 18)
Entropy (8)	Graph 26 (spread)
Chance shuffle and excess (9–11)	H1, H2 (Graphs 8, 16); H3 check (Graph 17)
p-value, Holm, BH (12–13)	Every test result
Error bars (14)	Every bar chart with whiskers
Group comparison (15)	H3 (Graphs 9, 17)
Other tools (16)	Graphs 25, 26, 28

Part 2: The study in one page

The question

When the words of a text stay exactly the same, but we tell the reader who wrote it, does the emotion they read into it change? And do large language models (LLMs) react the way people do?

Take the sentence *"I can't believe I actually did that."* It could be pride, embarrassment or regret, depending on who you imagine saying it. This study measures that effect.

What was done

- **300 short texts** were taken from the crowd-enVent corpus, where people described an emotional experience. They were split into three groups of 100:
 - **Unambiguous:** readers clearly agree on the emotion (the control group).
 - **Author-independent ambiguous:** readers disagree, but for reasons unrelated to who wrote it (e.g. missing details).
 - **Author-relevant ambiguous:** readers disagree, and knowing the author could plausibly change the reading.
- Every text was shown **without an author** (neutral) and with **two different personas** (persona A and persona B, chosen per text, e.g. "a first-time mother" vs "an experienced mother").
- **Humans** rated each version (about 3 people per version; no person ever saw two versions of the same text).
- **7 LLMs** (OpenAI, Claude, Gemini, Qwen3, Gemma3, Ministral, Llama 3.1) rated every version under **6 prompt set-ups** (zero/one/few-shot × numbered/inline list). The models also got a fourth version: "written by a person" with no specific persona.
- Everyone chose one emotion from the same 11: anger, disgust, fear, guilt, joy, pride, relief, sadness, shame, surprise, trust.

Research questions

	Question
RQ1	Does perceived author identity change human emotion judgments of ambiguous texts?
RQ2	Does perceived author identity change LLM emotion classifications?
RQ3	Do LLMs react like humans, in direction and in size ?

Hypotheses (exact wording from the proposal)

	Wording	In plain words
H1	Human emotion judgments will differ significantly across author-framing conditions for ambiguous texts, but not (or substantially less) for unambiguous control texts.	People read ambiguous texts differently when told who wrote them; clear texts should hardly change.
H2	LLM emotion classifications will shift across author-framing conditions, indicating sensitivity to minimal author cues.	Adding a short "Author: ..." line changes what the LLM answers.

	Wording	In plain words
H3	Framing effects will be stronger for author-relevant ambiguous texts than for author-independent ambiguous texts.	The author matters most when the ambiguity is about who is speaking.
H4	The direction of framing-induced shifts in LLM classifications will partially align with human judgment shifts, but the magnitude of these shifts will differ.	LLMs move somewhat towards the same emotions as people, but by a different amount.

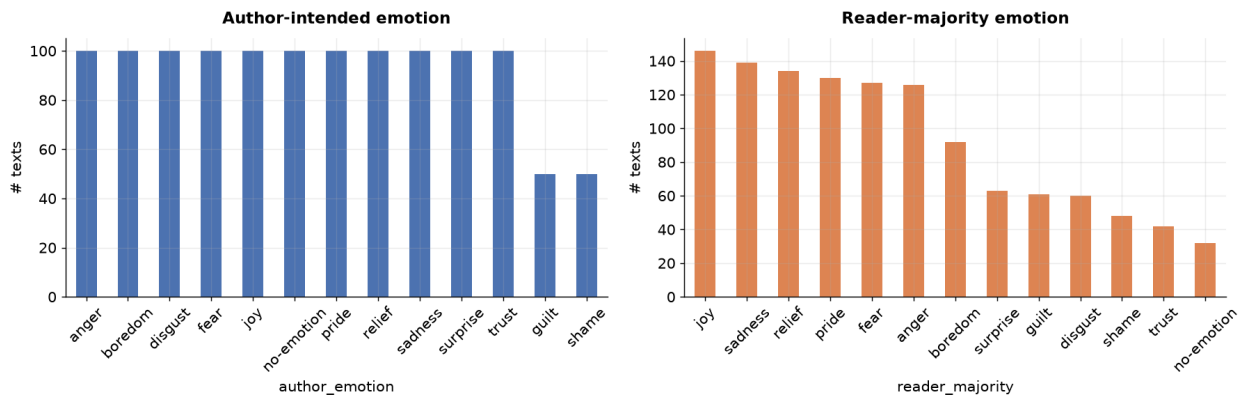
RQ1 is answered by H1 (and refined by H3), RQ2 by H2, and RQ3 by H4.

Part 3: Notebook 00, corpus and sampling

This notebook tests no hypothesis. It explains **how the 300 texts were chosen and grouped**, because every later result depends on those choices.

The source: crowd-enVent (Troiano et al., 2023). A subset of **1,200 texts** each has the emotion the **author** intended and the emotions chosen by **5 independent readers** (6,000 votes). Readers recover the author's own emotion for only **58.1%** of these texts.

Graph 1. What authors meant vs. what readers picked



What the graph shows

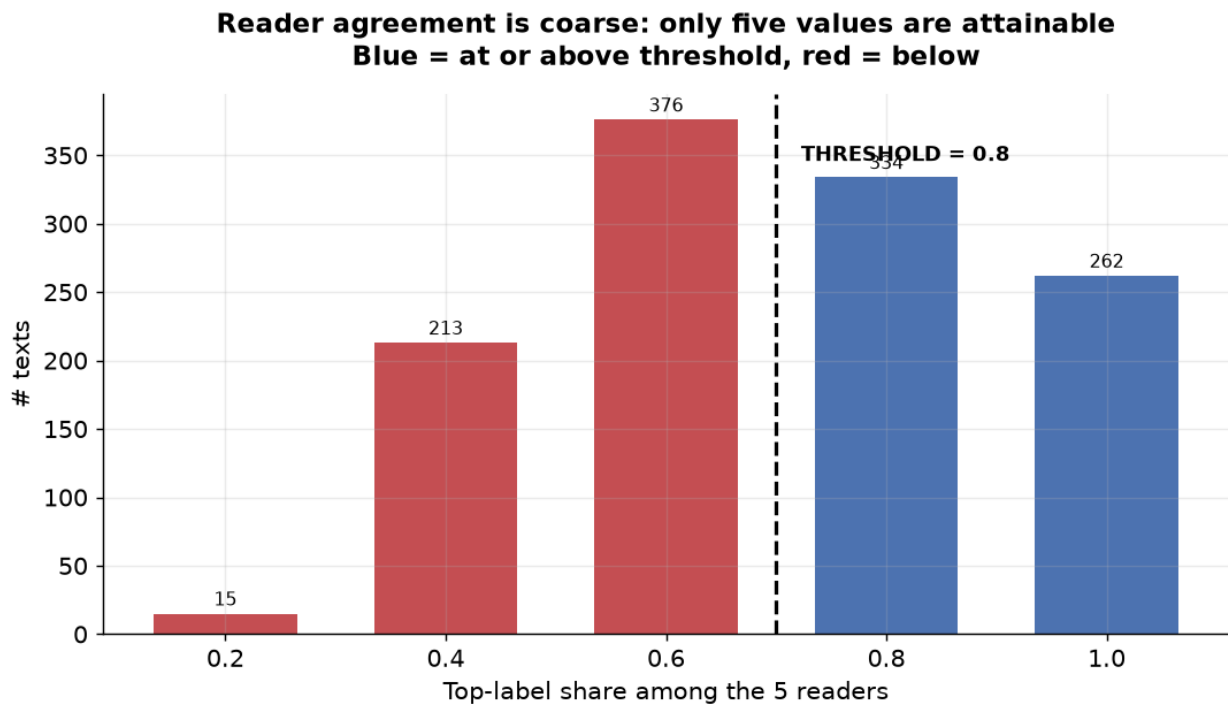
- Left: the emotion the authors said they felt. It is flat (100 texts each; guilt and shame 50) because the corpus was collected that way.
- Right: the emotion most readers picked for the same texts. It is uneven: joy, sadness and relief are picked far more often; trust and shame far less.

What it means

- Readers and authors do not simply agree. Disagreement is built into the data before any persona is added.
- The corpus also contains "boredom" and "no-emotion", which are outside the 11-emotion set. Those texts are removed later.

Say it like this: Before we change anything, readers already disagree with the authors: the two bar charts have different shapes.

Graph 2. How strongly do the 5 readers agree?



What the graph shows

- Each bar counts texts by the share of readers who chose the most popular emotion. With 5 readers only five values are possible: 1/5, 2/5, 3/5, 4/5, 5/5.
- The most common outcome is **3 of 5** (376 texts). 3 of 5 is already a majority: **972 of the 1,200 texts** (3/5, 4/5 or 5/5) have a strict majority. Only 262 are unanimous.
- The dashed line is the threshold: **4 of 5 readers or more** (blue) can count as unambiguous.

What it means

- Most texts have a majority, but it is rarely unanimous: in 938 of 1,200 texts at least one reader disagrees, and 228 texts (1/5 or 2/5) have no majority at all.
- 134 of the 1,200 texts even have a tie at the top.
- Partial disagreement is the normal case. That is why the thesis keeps full vote distributions instead of one winning label.

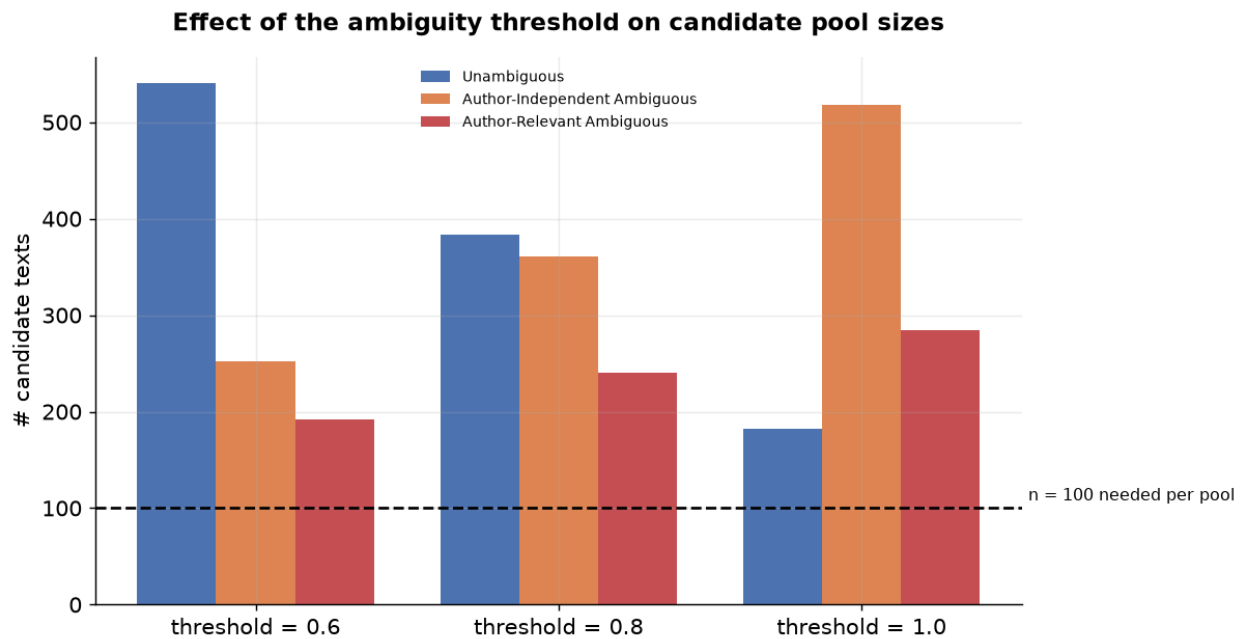
Say it like this: Most texts do have a majority, usually three out of five, but only about one in five is unanimous. Some disagreement is the normal case.

The grouping rule (three steps, always in this order)

- **Step 0:** drop texts whose author label or reader-majority label is boredom or no-emotion. 215 dropped, **985 remain**.
- **Step 1: Unambiguous** if at least 4 of 5 readers agree *and* they agree with the author.
- **Step 2:** for the rest, two personas were judged in a pilot. If the two personas gave **different** emotions, the text is **author-relevant**; if the same, **author-independent**.

The order matters: a text readers already agree on stays unambiguous, even if a persona could flip it. That protects the control group.

Graph 3. What if the threshold were different?



What the graph shows

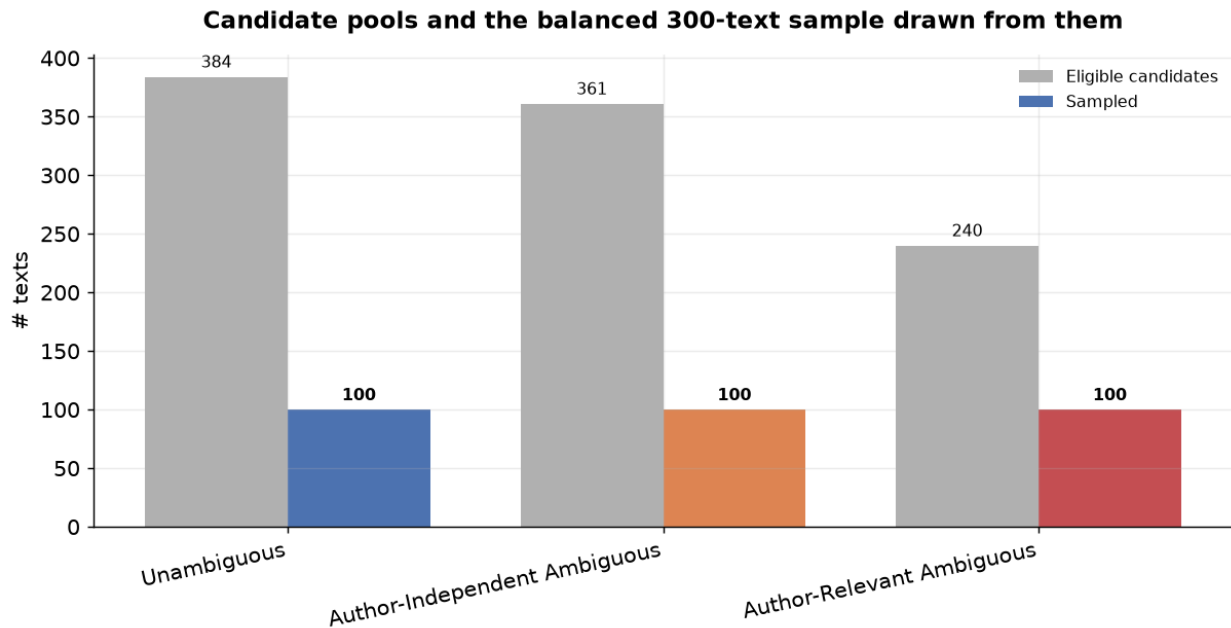
- Pool sizes for three possible thresholds: 3 of 5 readers (0.6), 4 of 5 (0.8, the one used), 5 of 5 (1.0).
- The dashed line is the 100 texts needed per group. Every setting has enough.

What it means

- 0.8 was not forced by a lack of texts. It was chosen because 3 of 5 is too weak for a control group, and 5 of 5 would allow only unanimous texts.
- It is a reasoned choice, not a statistically optimised one.

Say it like this: We needed 100 texts per group, and every threshold gave enough. We chose four out of five readers because it makes the control group reliably clear.

Graph 4. From candidates to the final 300



What the graph shows

- Grey: eligible candidates per group (384, 361, 240).
- Coloured: the 100 texts in the final sample for each group.

What it means

- The groups are exactly balanced, so group comparisons are not distorted by different sizes.
- Seven automatic checks passed: 300 texts, 100 per group, no duplicates, no prompt-example texts, and every text matches the rule.
- 27 individual reader votes (1.8%) were outside the 11 emotions; they are left out of the vote shares, so 275 of 300 texts keep all 5 original votes.
- **How the sample was built:** not one fresh random draw. An existing sample was kept, every text was re-classified under the rule above, over-full groups were trimmed to 100 (fixed seed) and gaps were filled from unused eligible texts. The rule is reproducible; the exact membership is recorded in `sampled_texts.csv`.

Say it like this: Three equal groups of one hundred. The sample was reconciled against an earlier sample under a fixed rule, then checked automatically.

One caveat to keep in mind (it matters for H3)

Author-relevant texts were *defined* by the pilot persona test flipping the emotion (their flip rate is 100% by construction; author-independent is 0%). So H3 asks whether a grouping built from pilot judgments **predicts** effects in new data. That is a fair question, but it is a **transfer test**, not an independent discovery.

Part 4: Notebook 01, from majority labels to distributions

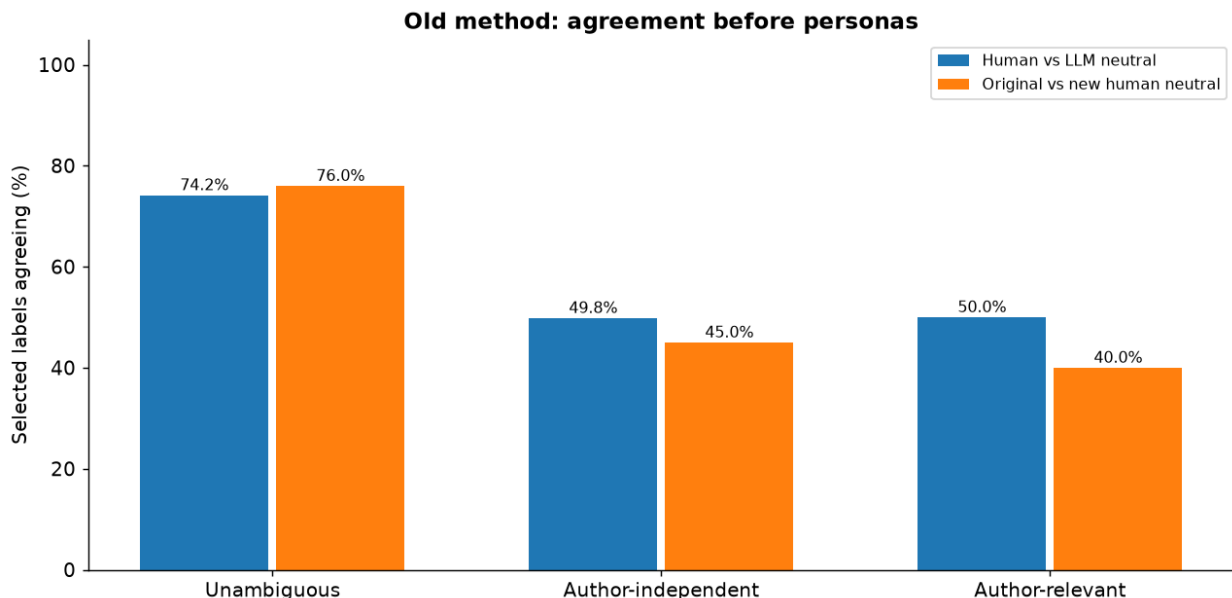
The notebook tells one story in four parts:

- **Part A:** analyse everything the **old way**, keeping only the winning emotion.
- **Part B:** show what that old way **throws away**.
- **Part C:** redo the same questions with **full vote distributions** (vectors).
- **Part D:** compare the two answers.

Part A: the old method (one label per text)

Each set of votes becomes one **selected label**, the most common emotion (ties broken alphabetically). Example: "surprise, joy, anger" (a three-way tie) becomes "anger", even though only 1 of 3 people chose it.

Graph 5. Before any persona: do labels agree?



What the graph shows

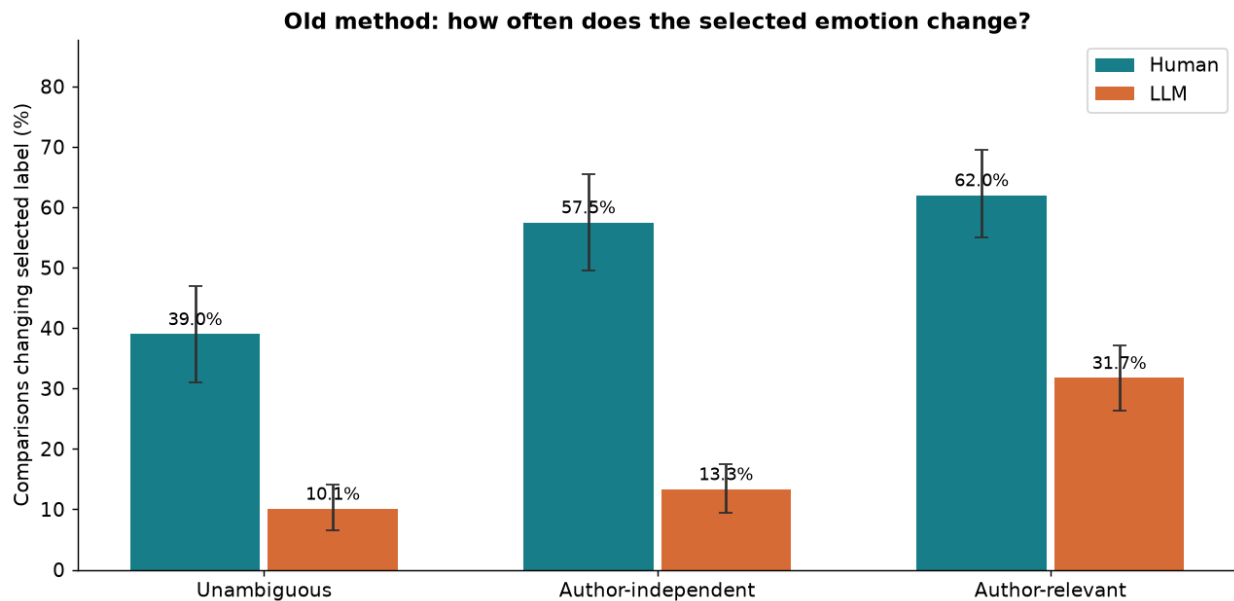
- Blue: how often humans and the LLM ensemble pick the same label with no author given.
- Orange: how often the **original** corpus readers and **our new** human raters pick the same label, again with no author.
- Unambiguous: about 75% agreement. Both ambiguous groups: only 40–50%.

What it means

- Two groups of people reading the **same text with no persona** already disagree about half the time on ambiguous texts.
- Any persona effect has to be judged against this background disagreement.

Say it like this: Even with no author at all, two groups of people pick the same emotion only about half the time on ambiguous texts. That is the noise we have to beat.

Graph 6. Old method: how often does the chosen emotion change?



What the graph shows

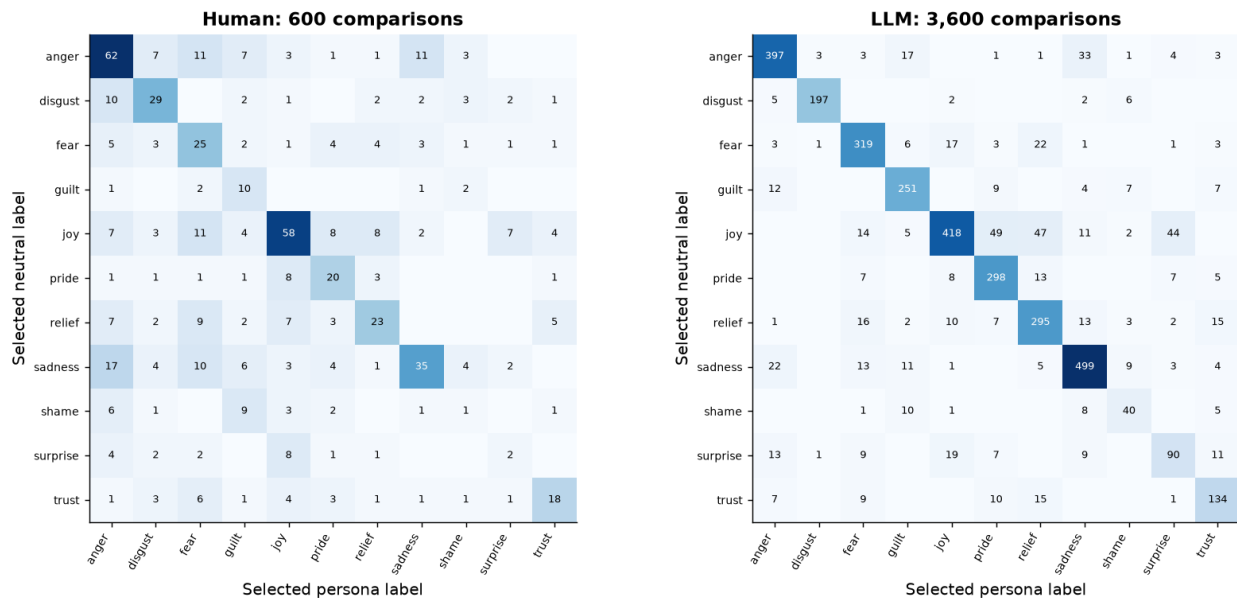
- Share of comparisons where the winning emotion under the persona differs from neutral.
- Humans: 39% (unambiguous), 58% (author-independent), 62% (author-relevant).
- LLMs: 10%, 13%, 32%. Overall: humans 52.8%, LLMs 18.4%.

What it means

- Both humans and LLMs change more often on ambiguous texts, and humans change far more often than LLMs.
- But this is not yet evidence. Humans change 39% of the time even on clear texts, partly just because different people rated each version.

Say it like this: The winning emotion changes more on ambiguous texts, and much more for people than for models. Whether that is more than chance comes next.

Graph 7. Which emotion turns into which?



What the graph shows

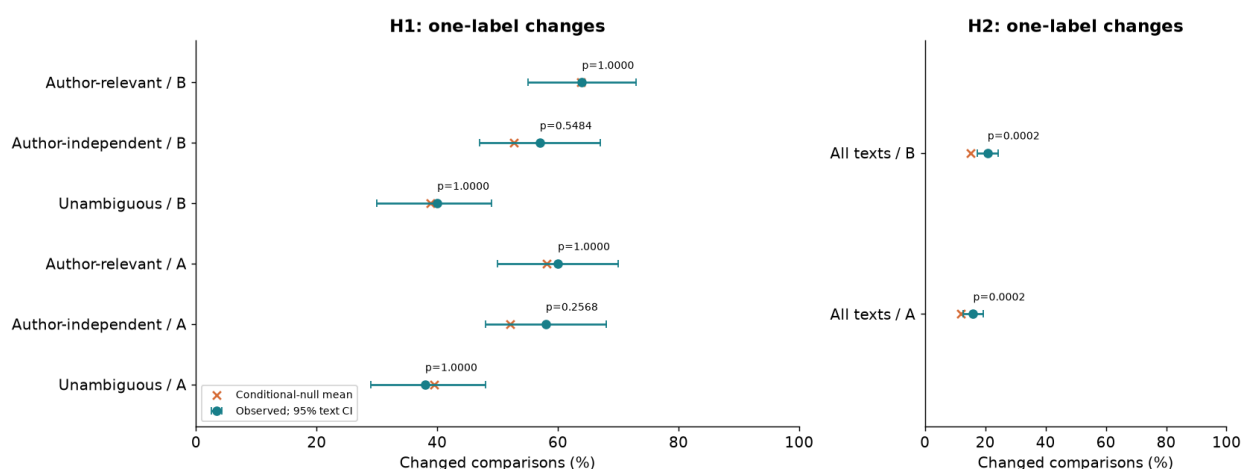
- Rows: winning emotion without a persona. Columns: winning emotion with a persona.
- The diagonal (top-left to bottom-right) = no change. Off-diagonal = a switch.
- Humans (600 comparisons) have many switches, e.g. sadness to anger (17). LLMs (3,600) stay mostly on the diagonal; their main switches are joy to pride (49), joy to relief (47) and joy to surprise (44).

What it means

- LLMs mainly move between neighbouring positive emotions. Humans switch more widely.
- The old method still cannot say *how much* moved; a switch is just yes or no.

Say it like this: Models mostly keep their answer, and when they switch, it is usually joy turning into pride, relief or surprise. People switch much more widely.

Graph 8. H1 and H2 tested the old way



What the graph shows

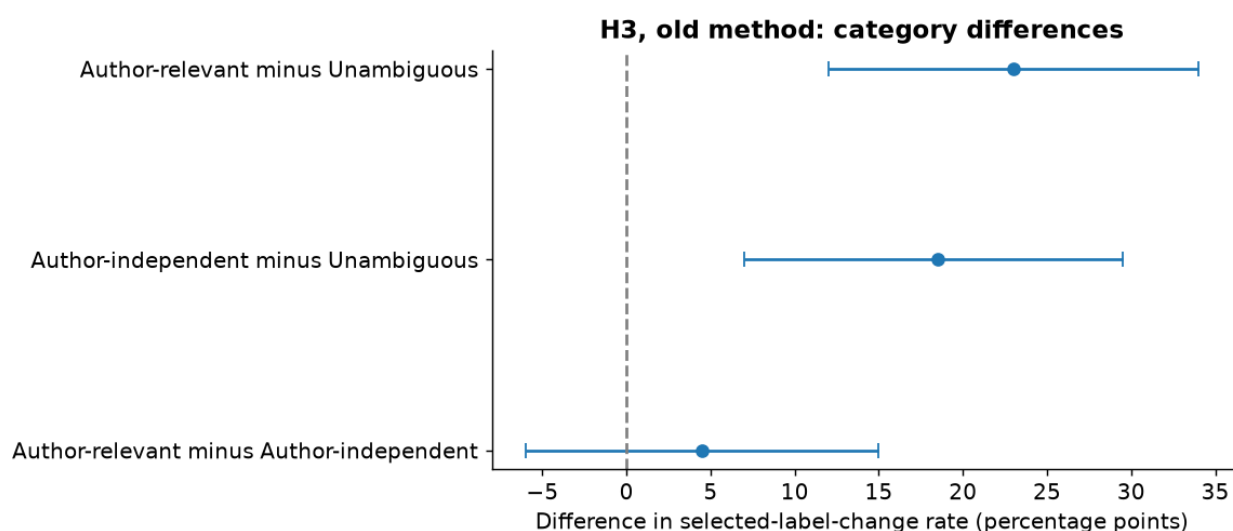
- Green dot = observed change rate (with its range). Orange × = what chance alone gives. The number is the corrected p-value.
- Left (humans, H1): every dot sits almost on its ×. **0 of 6 tests are significant.**
- Right (LLMs, H2): the dot sits clearly to the right of the ×. **Both tests significant** ($p = 0.0002$).

What it means

- With the old one-label method, **no human framing effect is detected at all.**
- The LLM effect is already detectable with the old method.
- This sets up the key argument: is there really no human effect, or is the old method blind to it?

Say it like this: Counting only the winning emotion, people look like they are not affected by the author at all. Models clearly are. Keep that in mind, because the new method changes the human answer.

Graph 9. H3 tested the old way



What the graph shows

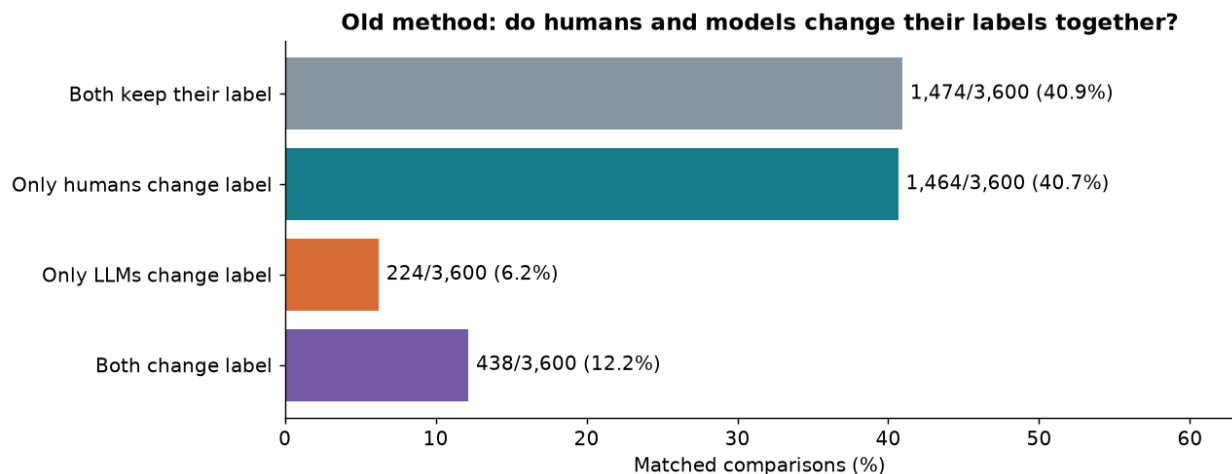
- Differences in label-change rate between groups. The dashed line is zero (no difference).
- Author-relevant vs. unambiguous: +23 points. Author-independent vs. unambiguous: +18.5. Both ranges are clear of zero.
- **Author-relevant vs. author-independent (the actual H3 question): +4.5 points, range -6 to +15, $p = 0.47$.** The range crosses zero.

What it means

- Ambiguous texts change more than clear ones.
- The old method cannot show that author-relevant texts change more than author-independent ones. H3 is not supported this way.

Say it like this: Ambiguous texts change more than clear ones, but between the two ambiguous groups the old method sees no real difference.

Graph 10. H4 tested the old way: do humans and models change together?



What the graph shows

- For each matched comparison: both keep their label (40.9%), only humans change (40.7%), only LLMs change (6.2%), both change (12.2%, 438 cases).
- In those 438 cases, humans and LLMs made **exactly the same switch** (e.g. both sadness to guilt) 23.3% of the time. Random pairing gives only 1.8% ($p = 0.0001$).

What it means

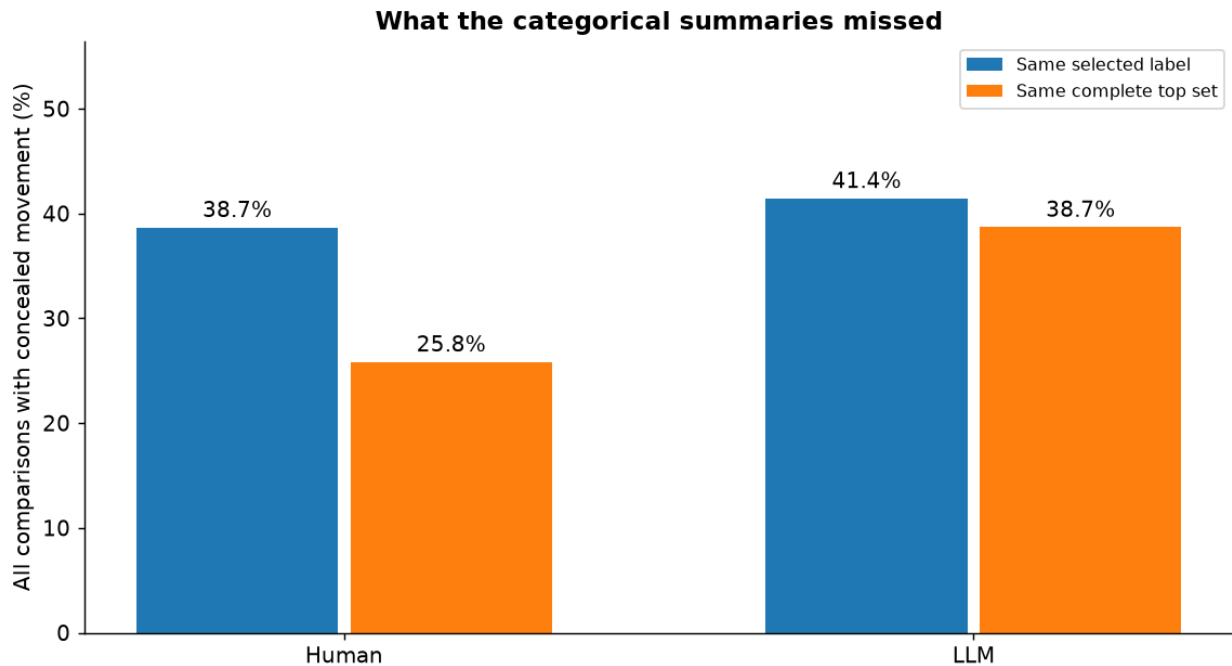
- Humans change far more often than models (a 34-point gap in change rate).
- When both change, they switch the same way much more often than chance.
- These are only rough stand-ins for H4: one label cannot measure the size or direction of a full shift.

Say it like this: People change their answer much more often than models, but when both change, they often make the same switch, far more than chance would.

Part B: what the old method throws away

Ties change the answer. In 262 of 600 human comparisons at least one side was tied. Just reversing the alphabetical tie rule changes the changed/unchanged decision in 125 comparisons and moves the human change rate from 52.8% to 57.3%. The result depends on an arbitrary rule.

Graph 11. Movement the old method cannot see



What the graph shows

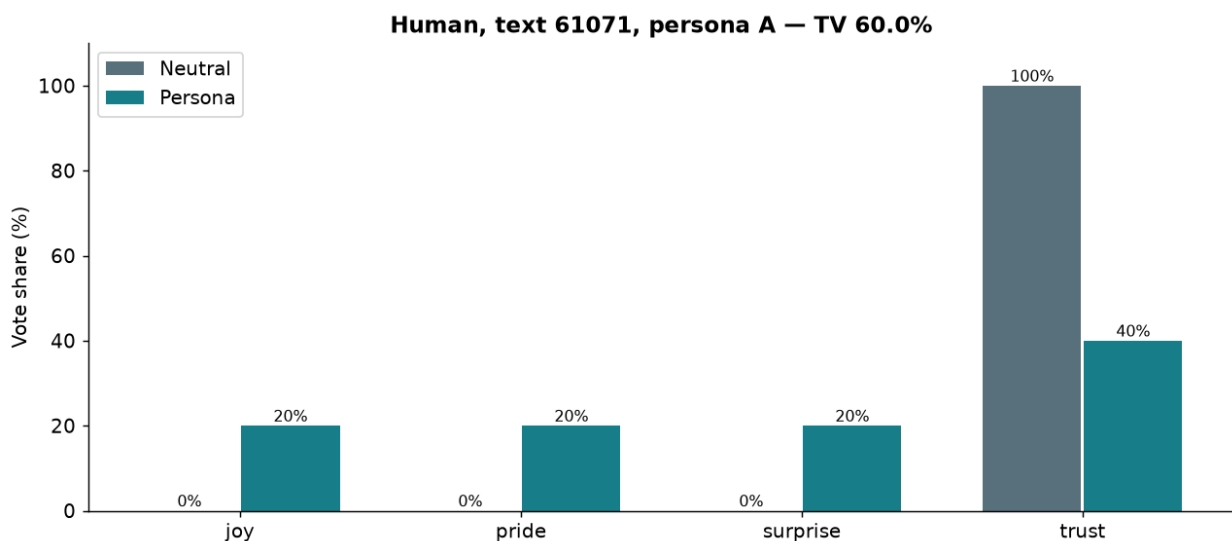
- Blue: comparisons where the **winning label stayed the same**, but the votes still moved. Humans 38.7% (232 of 600); LLMs 41.4% (1,492 of 3,600).
- Orange: the stricter version, where even the full set of tied winners stayed the same. Humans 25.8% (155); LLMs 38.7% (1,393).

What it means

- Roughly **a quarter to two-fifths** of all comparisons moved but would be recorded as "no change".
- For LLMs, most of the movement is invisible to a one-label analysis.

Say it like this: In about four out of ten comparisons, the vote moved but the winner stayed the same, so the old method recorded no change.

Graph 12. A real example of hidden change



What the graph shows

- Text: "My friend needed support to manage her money so she gave me control of her bank accounts." Persona: someone entrusted for the first time with this level of financial responsibility.
- Neutral: 100% trust (3 raters). Persona: 40% trust, plus 20% each for joy, pride and surprise (5 raters). Shift = 60%.

What it means

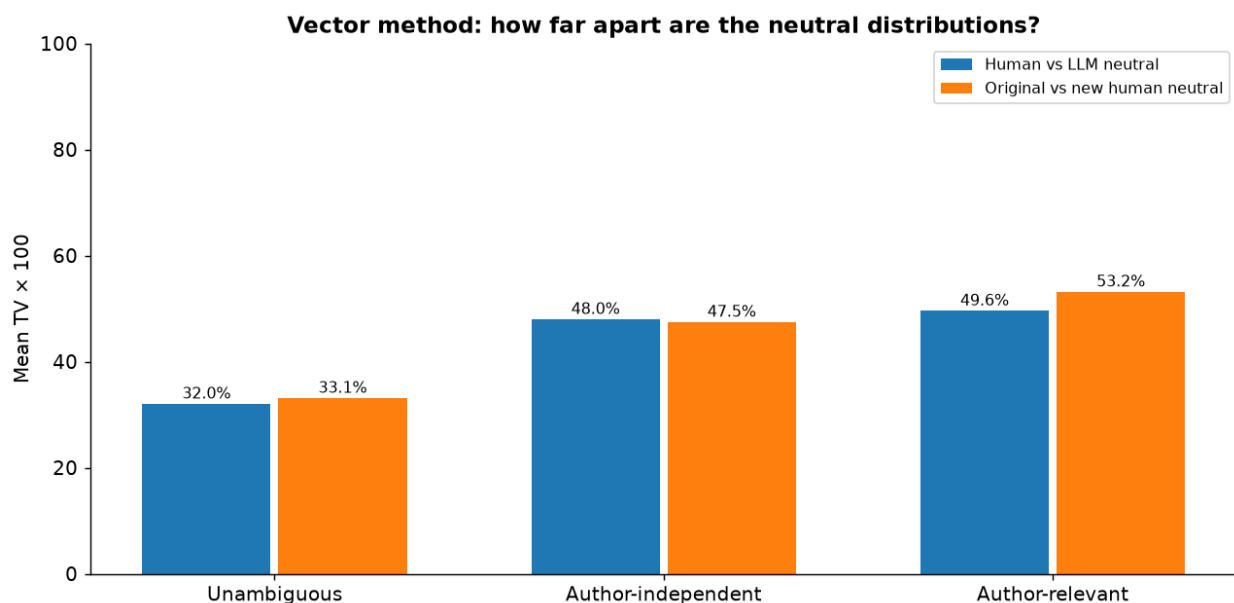
- The winning label is **trust in both cases**, so the old method says "no change".
- In reality, three new emotions appeared and trust lost 60 points.
- This example was chosen because it is the clearest one, not because it is typical.

Say it like this: The winner stays "trust", so the old method sees nothing. But the persona brought in joy, pride and surprise, and more than half the votes moved.

Part C: the vector method

Keep every vote as a percentage across all 11 emotions. Then measure the **shift**: how much of the vote share must move to get from neutral to persona. The texts, the people and the models stay exactly the same as in Part A; only the way the answers are summarised changes.

Graph 13. How far apart are two no-persona measurements?



What the graph shows

- The same comparisons as Graph 5, now measured as shift instead of match or no match.
- Original readers vs. new readers, both with no persona: 33% (unambiguous), 48% and 53% (ambiguous groups).

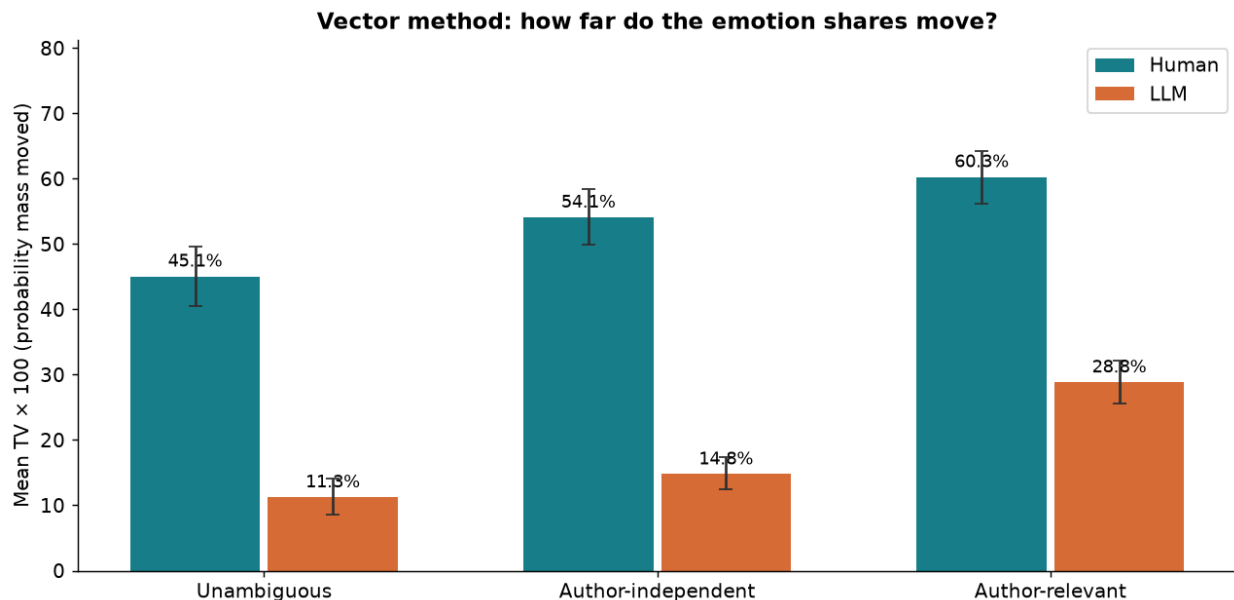
What it means

- Two independent groups of people reading the same text with no persona differ by a third to a half of their votes.
- This shows how much **background disagreement** exists between rater groups. It is context, not a number to subtract: the two rounds also differ in raters and setting. A raw human shift of 45% on clear

texts is therefore not impressive on its own; it has to beat the chance reference in the tests.

Say it like this: With no persona at all, two groups of readers still differ by a third to a half of their votes. Every persona effect has to be judged against that.

Graph 14. How far do the votes move under a persona?



What the graph shows

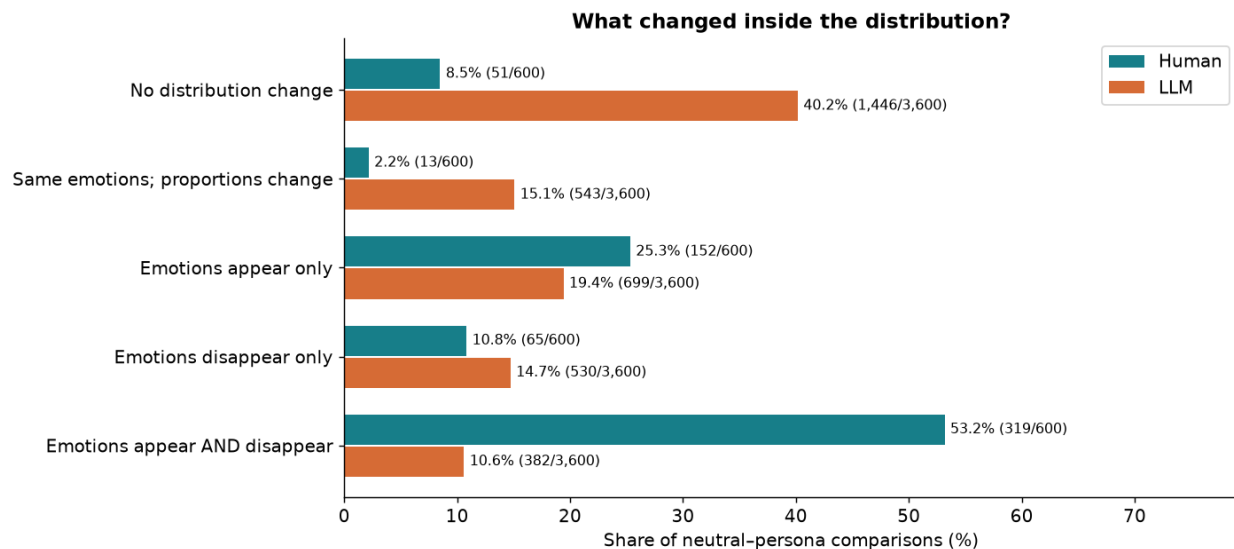
- Average shift. Humans: 45.1% (unambiguous), 54.1% (author-independent), 60.3% (author-relevant). LLMs: 11.3%, 14.8%, 28.8%.
- Overall: humans 53.2%, LLMs 18.3%.

What it means

- Same ordering as the old method, but now the size is measured, not just yes or no.
- The human numbers still include ordinary rater-to-rater disagreement (Graph 13); the tests that account for it follow.

Say it like this: Under a persona, people's votes move about half-way on average, the models' about a fifth. Both move most on author-relevant texts.

Graph 15. What kind of change happened?



What the graph shows

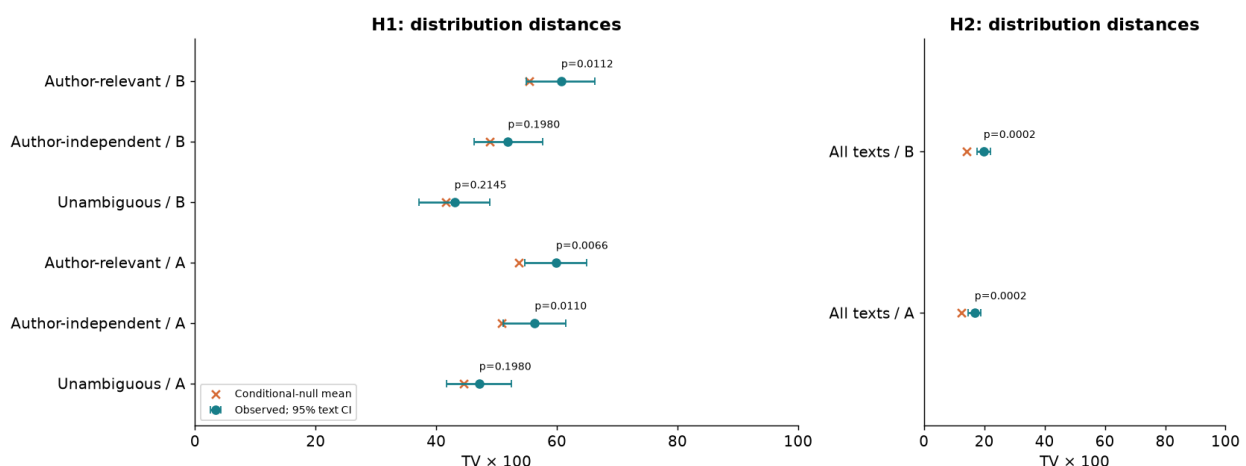
- Every comparison falls into exactly one of five types: no change at all; same emotions but different percentages; new emotions appear; emotions disappear; both.
- Humans: only 8.5% show no change; 53.2% have emotions both appearing and disappearing.
- LLMs: 40.2% show no change; 15.1% keep the same emotions but change the percentages.

What it means

- Humans mostly **swap in new readings**. Models more often stay put or **re-balance** the same emotions.
- These are statements about the **group's vote distribution**, not about individual people changing their minds: different people rated each version.

Say it like this: People tend to bring in new emotions and drop old ones. Models more often keep the same emotions and just shift the balance, or do not change at all.

Graph 16. H1 and H2 tested with vectors (the main result)



What the graph shows

- Same layout as Graph 8: green dot = observed shift, orange x = chance.

- **H1 (humans):** author-relevant A ($p = 0.0066$) and B ($p = 0.0112$), and author-independent A ($p = 0.0110$) are significant. Author-independent B ($p = 0.198$) and both unambiguous tests ($p = 0.198, 0.2145$) are not.
- **H2 (LLMs):** both significant ($p = 0.0002$).

What it means

- The old method found **0 of 6** human effects; the vector method finds **3 of 6**. The vector method detects human framing effects that the one-label test did not.
- Unambiguous texts show nothing detectable, as H1 predicted.
- LLMs are clearly affected under both personas.

Say it like this: With full distributions, the human framing effect appears: clearly for author-relevant texts, for one persona on author-independent texts, and none detected for clear texts. That is broadly the pattern H1 predicted.

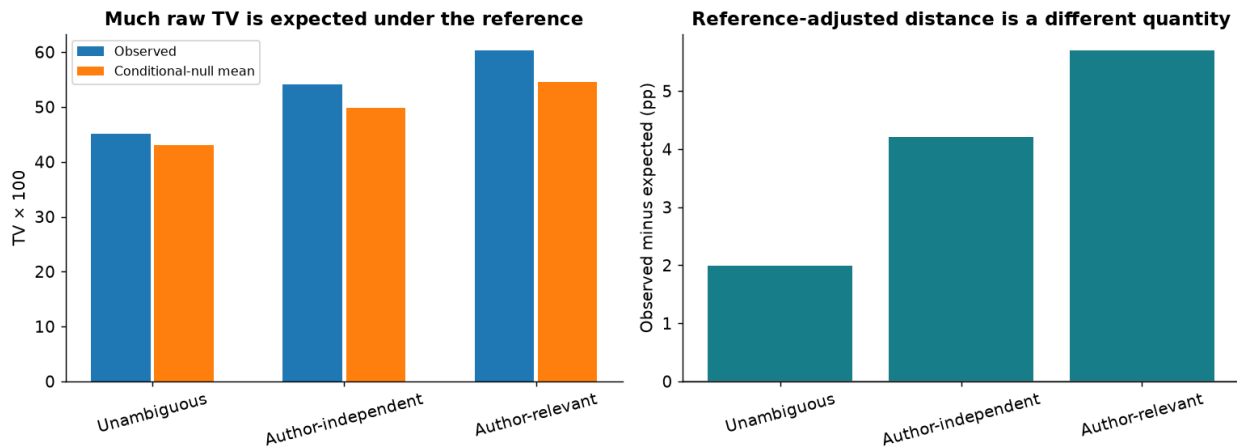
Why does author-independent pass for persona A but not for persona B?

Author-indep.	Observed shift	Chance gives	Extra shift	p	Passes?
Persona A	56.3%	50.9%	+5.4 points	0.011	Yes
Persona B	51.9%	48.9%	+3.0 points	0.198	No

- Both extra shifts are **positive**: both personas pushed readings in the same direction. The effect is small and the two results fall either side of the cut-off.
- A and B are not two types of persona; they are two different personas per text (e.g. "a spouse" vs "a girlfriend"), rated by different people. Some difference between them is expected.
- **A and B were never tested against each other.** So the data do not show that persona A had a bigger effect than persona B. Their observed ranges overlap a lot (A: 51–62%, B: 46–58%), which gives no sign of a real difference, but overlap alone is not a test. One result passing and the other not is not, by itself, evidence that they differ.
- It fits a gradient: author-relevant passes twice (+6.1, +5.3), author-independent sits at the edge (+5.4, +3.0), unambiguous passes neither (+2.6, +1.4).

Say it like this: For author-independent texts the effect is smaller and sits right at the detection threshold. It reaches significance for one persona set and not the other. That alone does not show the two persona sets had different effects; we did not test them against each other.

Graph 17. H3 tested with vectors: raw vs. chance-adjusted



What the graph shows

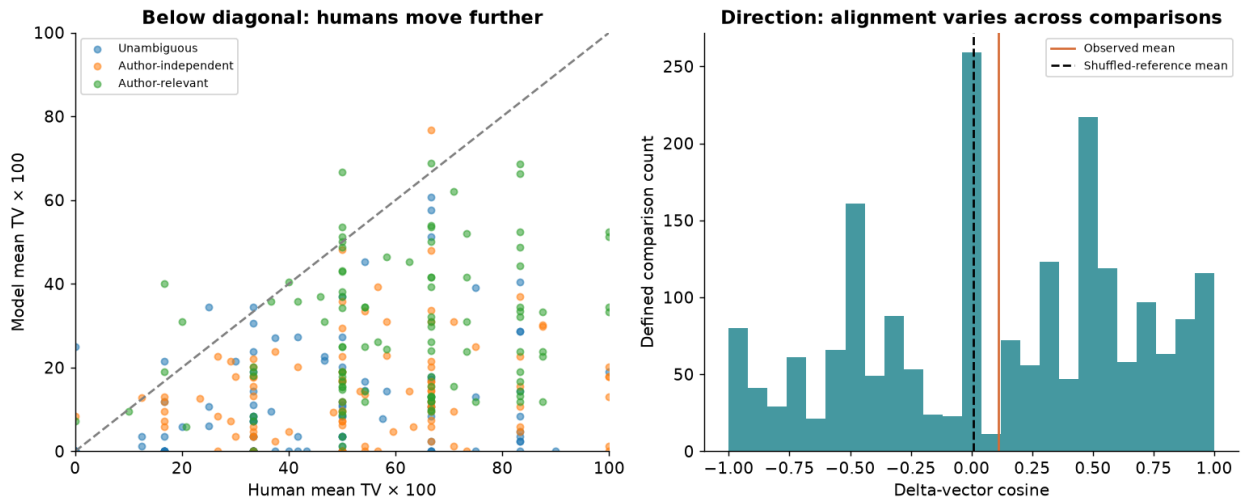
- Left: observed shift (blue) next to the shift chance alone would give for those same texts (orange). Chance already gives 43%, 50% and 55%.
- Right: what is left after subtracting chance: +2.0, +4.2 and +5.7 points.
- **Raw test** (author-relevant vs. author-independent): 6.2 points, $p = 0.040$, significant.
- **Chance-adjusted test**: 1.5 points, $p = 0.70$, not significant.

What it means

- Part of the raw difference **reflects how much these texts' raters disagree to begin with**: three people who naturally disagree will give different votes every time. The adjusted figure is a reference-based sensitivity check, not the "true" framing effect.
- The ordering is still there after adjusting ($2.0 < 4.2 < 5.7$), but the differences become too small to separate from chance with 100 texts per group.
- The raw test is the main (planned) comparison. The adjusted test is a **post-hoc sensitivity analysis**, added after seeing the raw result, so it is reported next to the raw test, not instead of it. "Not significant" does not prove the groups are equal.

Say it like this: In the main test, author-relevant texts shift more, and it is significant. But in a sensitivity check that allows for how disputed each text already is, the difference is too small to confirm. So the H3 result depends on the reference we use.

Graph 18. H4 tested with vectors: size and direction



What the graph shows

- **Left (size):** each dot is a text. Dots below the dashed diagonal mean humans shifted more than models. Almost all dots are below it. Average gap: **34.9 points** (range 32.2–37.5).
- **Right (direction):** for each comparison, how similar the human and model shifts are in direction (+1 same, 0 unrelated, -1 opposite). The spread is wide, but the average is **0.11** (orange line), against about 0 for random pairing (black line), $p = 0.0001$.
- Direction can only be measured when both sides moved: 2,020 of 3,600 comparisons. The other 1,580 are left out, not counted as zero.

What it means

- Size: humans shift much more than models. H4 predicted the sizes would **differ**, so this supports it.
- Direction: models lean towards the same emotions as people more than chance, but only slightly. H4 predicted **partial** alignment, so this is consistent with it.
- Qualifications: direction is only defined where both sides moved, and the two sides were measured differently (different people per human version; the same model on both sides). So this is qualified support, not a clean confirmation.

Say it like this: People move much further than models. Models do lean in the same direction as people, more than chance, but only weakly. That is consistent with H4, with the caveats about how direction is measured.

Part D: old method vs. vector method

	Old method (one label)	Vector method
H1 humans	0 of 6 tests significant	3 of 6 significant (author-relevant A and B, author-independent A)
H2 LLMs	2 of 2 significant	2 of 2 significant
H3	AR vs AI: 4.5 points, $p = 0.47$	Raw: 6.2 points, $p = 0.040$. Chance-adjusted: 1.5 points, $p = 0.70$
H4 size	Humans change label 34.4 points more often	Humans shift 34.9 points more

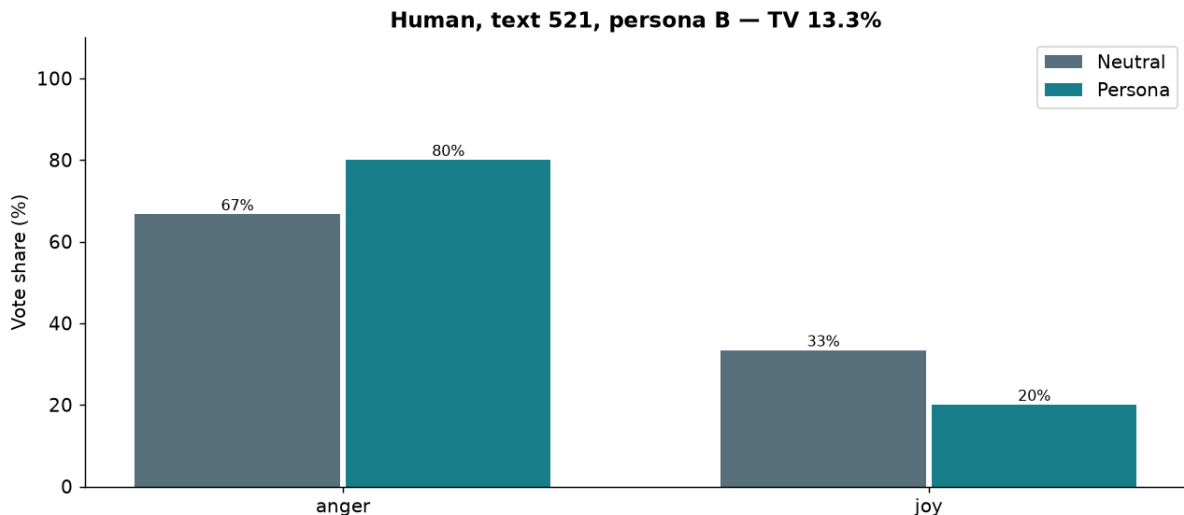
	Old method (one label)	Vector method
H4 direction	Same switch in 23.3% of joint changes (chance 1.8%)	Direction similarity 0.11 (chance about 0)
Hidden change	"Unchanged" means unchanged	26–41% of all comparisons moved while the label stayed the same. Among the comparisons the old method calls unchanged, 82% (humans) and 51% (LLMs) moved

Say it like this: A majority label only tells us the winner. Keeping the full vote distribution shows how far and where the votes move. It detects human framing effects the old method did not, and it also shows where an apparently strong result, H3, is not robust.

Part 5: Notebook 02, what changed inside the distributions

Notebook 01 answered the hypotheses. Notebook 02 looks closer at *what* moved. Its first graph repeats Graph 15 (the five change types), so it is not shown again.

Graph 19. Same emotions, different percentages: human example



What the graph shows

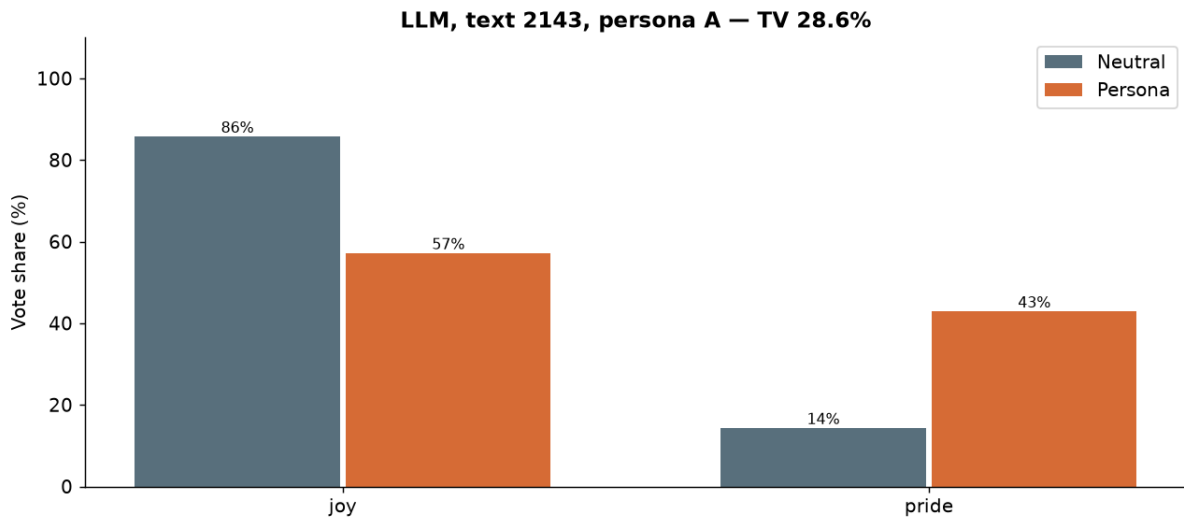
- Text: "A child that I was looking after was refusing to get off the bathroom floor and she needed to go to bed. Whenever I intervened she pushed me away." Persona: a grandparent.
- Anger goes from 67% to 80%, joy from 33% to 20%. Same two emotions, same winner.

What it means

- This is change the old method cannot see: the balance shifted but the label did not.
- For humans this pattern is rare: 13 of 600 comparisons (2.2%). Only 3 keep even the same set of tied winners.

Say it like this: Same emotions, same winner, but anger got stronger. A one-label summary would say nothing changed.

Graph 20. Same emotions, different percentages: LLM example



What the graph shows

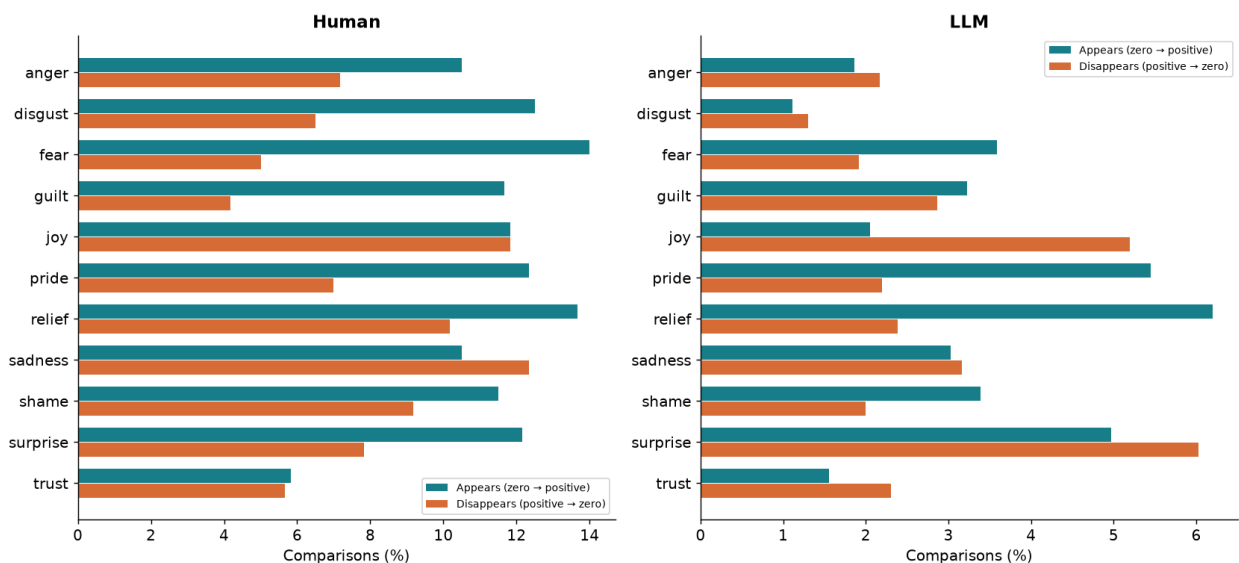
- Text: "I baked a delicious strawberry cobbler." Persona: a home baker (few-shot, inline prompt).
- Among the 7 models, joy drops from 86% to 57% and pride rises from 14% to 43%. That means two models switched from joy to pride.

What it means

- For models this pattern is common: 543 of 3,600 comparisons (15.1%), in 153 of the 300 texts. 310 (8.6%) keep even the same set of winners.
- Models often respond to a persona by re-weighting, not by switching.

Say it like this: Joy still wins, but two models switched to pride once the author was a home baker. Models often respond like this: they shift the balance without changing the winner.

Graph 21. Which emotions appear and disappear?



What the graph shows

- For each emotion: how often it **appears** (had no votes, then gets some) and **disappears** (had votes, then none), out of all comparisons.

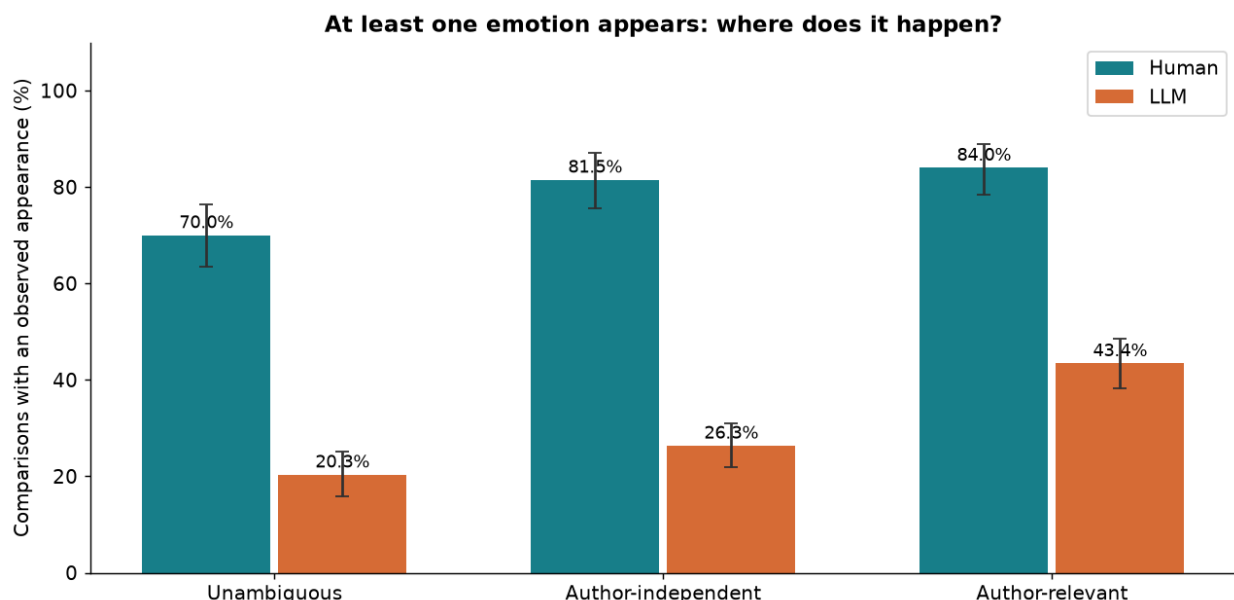
- Humans: fear appears most (14.0%). Guilt appears 11.7% but disappears only 4.2%. Sadness disappears slightly more than it appears.
- LLMs: relief (6.2%) and pride (5.4%) appear most. Joy disappears more than it appears (5.2% vs 2.1%).

What it means

- For models, personas tend to replace plain **joy** with more specific positive emotions such as **pride** and **relief**.
- For humans, personas tend to bring in **fear** and **guilt**.
- These are descriptive rates, not tested hypotheses.
- "Appears" means the emotion had no votes in one group's distribution and some in the other. It does not mean an individual person switched emotions.

Say it like this: When a persona is added, models tend to swap joy for pride or relief, and people tend to bring in fear and guilt.

Graph 22. Where do new emotions appear?



What the graph shows

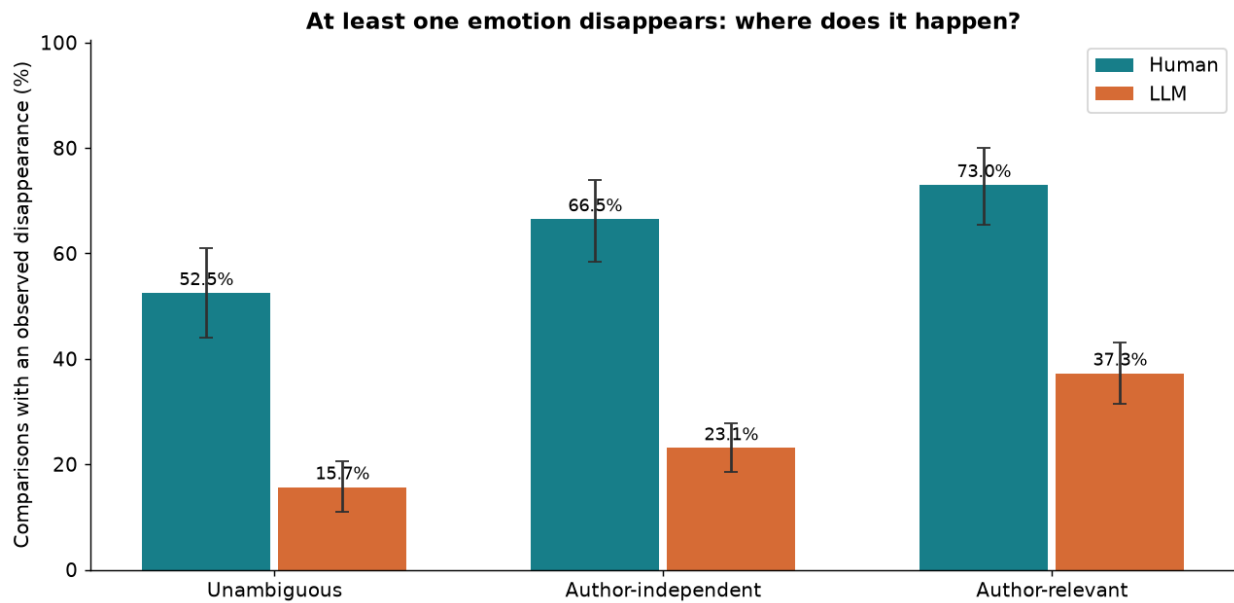
- Share of comparisons where at least one new emotion appears.
- Humans: 70.0%, 81.5%, 84.0%. LLMs: 20.3%, 26.3%, 43.4%.

What it means

- Both rise from clear to author-relevant texts.
- Human rates are much higher. This is a group-level statement: each version was rated by different people, and with only about 3 votes, one rater choosing a different emotion is enough to add one to the distribution.

Say it like this: New emotions appear most on author-relevant texts, for people and for models alike.

Graph 23. Where do emotions disappear?



What the graph shows

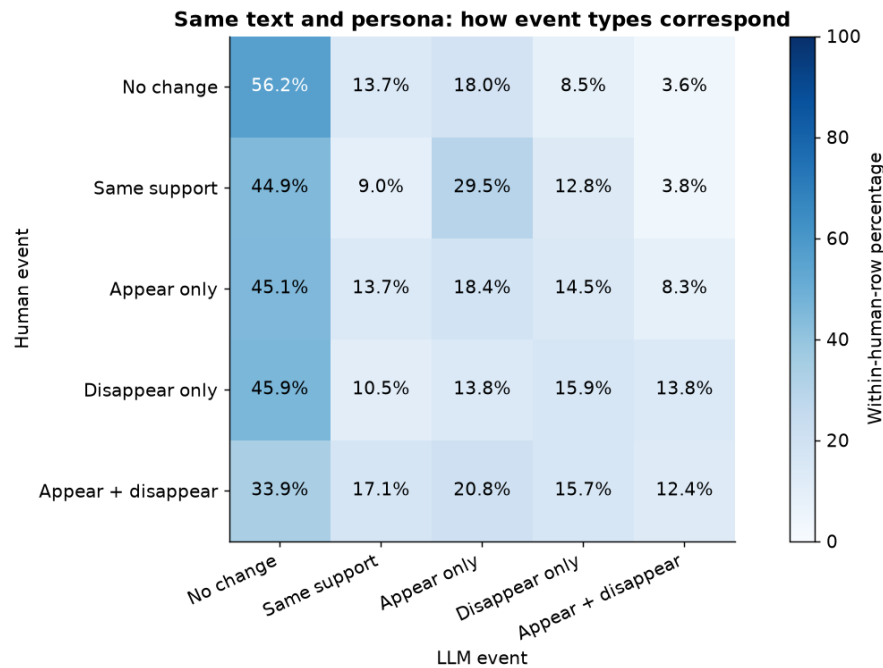
- Share of comparisons where at least one emotion disappears.
- Humans: 52.5%, 66.5%, 73.0%. LLMs: 15.7%, 23.1%, 37.3%.

What it means

- Same ordering as appearances. Many comparisons show emotions both leaving and entering the distribution, so the mix changes rather than simply growing.
- Like Graph 22, this describes a pattern; it is not a separate test of H3.

Say it like this: Emotions also drop out of the vote distribution most often on author-relevant texts, so the mix of readings changes rather than simply growing.

Graph 24. Do humans and models change in the same way?



What the graph shows

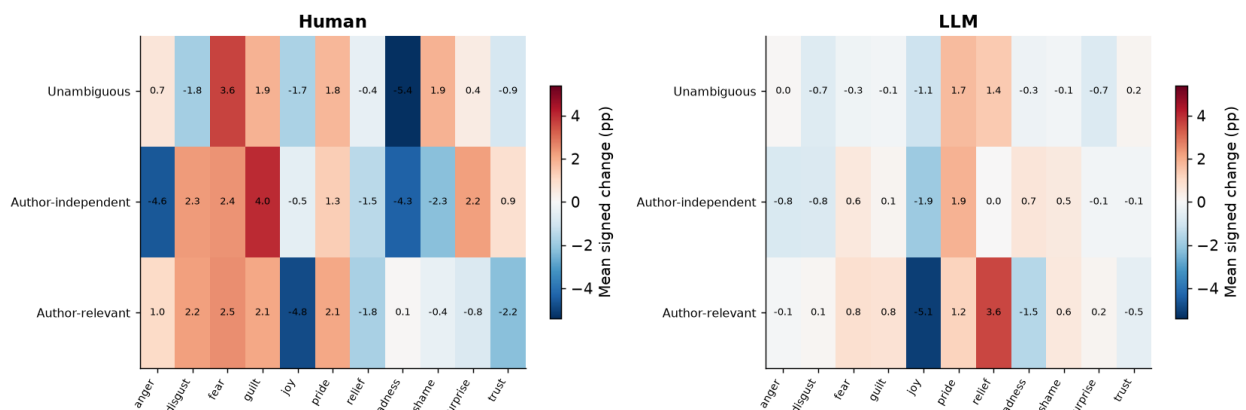
- Rows: what humans did for a text and persona. Columns: what the models did for the same text and persona. Each row adds up to 100%.
- When humans did not change (top row), models also did not change 56.2% of the time.
- In every row the most common model response is "no change" (34–56%).

What it means

- There is some matching (the top-left cell is the strongest), but it is weak: whatever people do, models most often stay put.
- This matrix is **descriptive**: no statistical test of agreement was run on it. It fits with the weak direction result in Graph 18, which was tested.

Say it like this: When people do not change, models usually do not either. But when people do change, models still most often stay put. Any correspondence here looks weak, and it was not formally tested.

Graph 25. Which emotions gain or lose on average?



What the graph shows

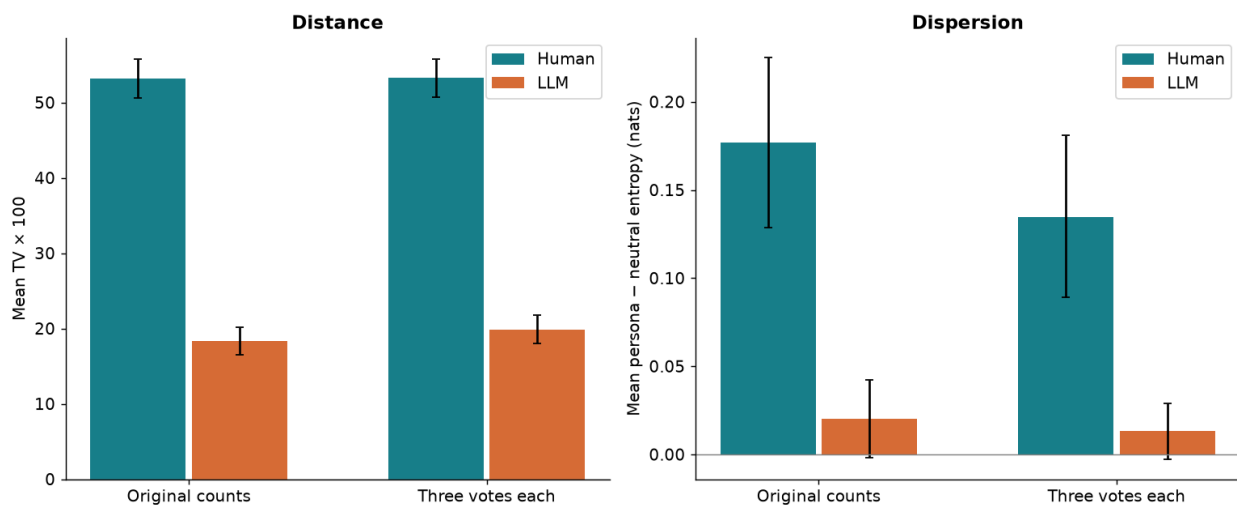
- Average change in each emotion's share (percentage points). Red = gained under the persona, blue = lost.
- 66 small tests (2 sources × 3 groups × 11 emotions), corrected with BH. **7 of 66 are significant**:
- Humans: guilt +4.0 on author-independent texts ($p = 0.041$).
- LLMs: pride +1.7 and disgust -0.7 (unambiguous); pride +1.9 and joy -1.9 (author-independent); joy -5.1 and relief +3.6 (author-relevant).

What it means

- Most colours are **not** significant, for example human sadness -5.4 on clear texts. Averages can also hide opposite shifts that cancel out.
- The significant model shifts all point the same way: away from joy, towards pride and relief. These are exploratory tests.
- **Note for the thesis:** the current text says no per-emotion shift survives correction (0 of 33). With this analysis, 7 of 66 do, including one human effect (guilt). That sentence needs updating.

Say it like this: Most per-emotion changes are too small to confirm. The clearest ones are the models moving from joy to pride and relief, and people moving towards guilt on author-independent texts.

Graph 26. Is the human-model gap just about vote counts?



What the graph shows

- Humans have about 3 votes per version, models up to 7, and fewer votes can make shifts look bigger. So both sides were cut to **3 votes each** (the same 3 models on both sides), 200 times.
- Left (shift): humans 53.2% to 53.2%, models 18.3% to 19.9%. Gap 34.9 to **33.4 points**.
- Right (spread / entropy): humans rise by 0.18 (0.13 after equalising); the range stays above zero. Models: 0.02 (0.01), and the range touches zero.

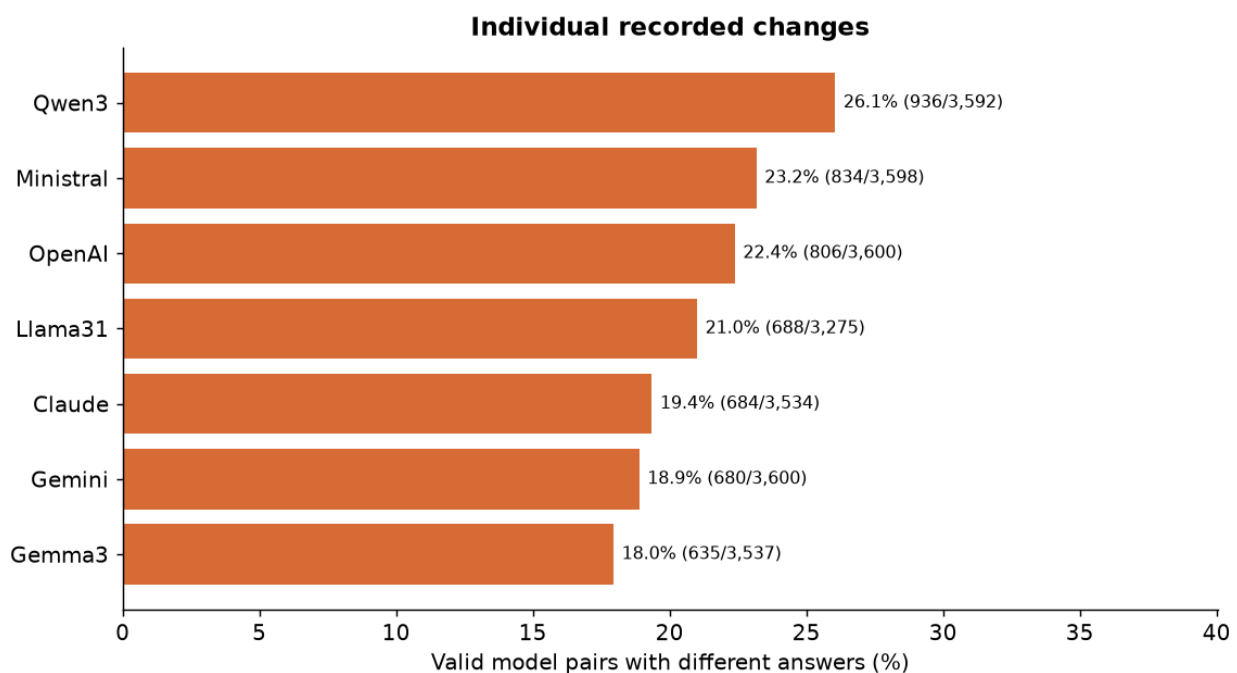
What it means

- Equalising vote counts barely changes the gap, so unequal vote counts do **not plausibly explain** it. This is a sensitivity check; it does not identify what causes the gap.

- Under a persona, human votes **spread out more** (the raters of a version agree less with each other). The model votes show no detectable increase in spread: their distribution moves without spreading out.
- Remaining caveat: different people rated each human version, while each model answered both versions itself.

Say it like this: Even with the same number of votes on both sides, people shift far more than models. Human votes also spread out under a persona; the model votes shift without detectably spreading out.

Graph 27. Does every model react?



What the graph shows

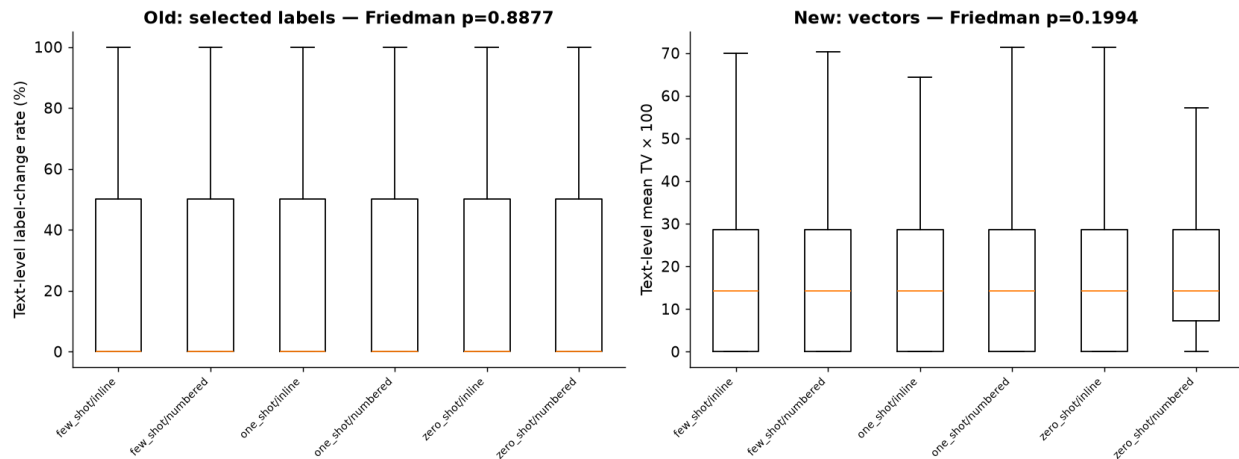
- How often each model gave a different answer under the persona than without one.
- From Gemma3 at 18.0% to Qwen3 at 26.1%. Overall 5,263 of 24,736 model answers changed (21.3%).

What it means

- Every model changed some of its answers, not just one or two models.
- These are **descriptive** rates from one answer per condition. No test compared the models, so we cannot say one model is statistically more sensitive than another.
- Note: in 61.2% of cases at most one model changed. The H2 test only picks up changes several models share, which is why these individual rates are reported next to it.

Say it like this: All seven models changed some answers when an author was added, between about one in five and one in four times. These are descriptive numbers, not a ranking.

Graph 28. Does the prompt set-up matter?



What the graph shows

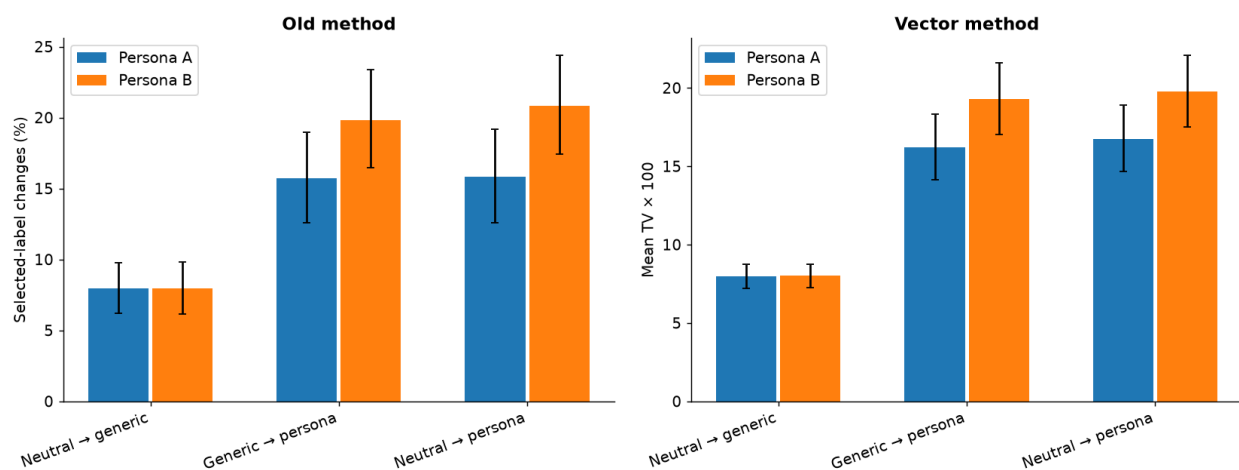
- Each box shows how much texts shifted under one of the 6 prompt set-ups. Left: old method (label change). Right: vector method (shift).
- All boxes look nearly identical. Tests: $p = 0.888$ (old) and $p = 0.199$ (vector).

What it means

- The prompt format (zero, one or few examples; numbered or inline list) makes **no detectable difference** to the persona effect.
- So the LLM effect does not appear to depend on one particular prompt. "No detected difference" is not proof of no difference: small differences could be missed.
- This is exploratory: the proposal has no hypothesis about prompt format.

Say it like this: We found no statistically detectable difference between the six prompt set-ups: the persona effect looks similar under all of them.

Graph 29. Is it the specific persona, or just naming any author?



What the graph shows

- Models only. Three steps: neutral to "written by a person" (generic), generic to the specific persona, and neutral to persona.

- Vector method: neutral to generic $\approx 8\%$; generic to persona $\approx 16\%$ (A) and 19% (B); neutral to persona $\approx 17\%$ and 20% . The old method gives the same pattern.

What it means

- Adding a generic author line is associated with a small shift. Changing from the generic author to the **specific persona** is associated with a shift about twice as large.
- This suggests the models are sensitive to **who** the author is, not only to **whether** an author is named.
- These three distances **do not add up** ($8 + 16$ is not 17), and should not be described as parts of the total effect or as percentages caused by each step.
- No generic-author version was collected for humans, so this is models only.

Say it like this: A generic author line is associated with a small shift. Going from that generic author to a specific person is associated with a shift about twice as big. So the models are sensitive to who the author is, not just to whether there is one.

One more check (no graph): H3 the old way, adjusted for chance

Applying the same chance adjustment to the old label method: author-relevant minus author-independent goes from $+4.5$ points ($p = 0.47$) to -4.1 points ($p = 0.48$). Neither method, once adjusted, shows author-relevant texts changing more than author-independent ones.

Part 6: The answers

How to read the verdicts

- **Supported:** every part of the prediction came true.
- **Partially supported:** some parts came true, others did not.
- **Weak / not robust:** the effect shows in the main test but not in a sensitivity check.
- **Qualified support:** the pattern is consistent with the prediction, with stated limits on what the measurement can show.

Research questions

	Answer
RQ1 humans	Yes, for ambiguous texts. Clearly for author-relevant texts (both personas); for one persona on author-independent texts; not detected for unambiguous texts.
RQ2 LLMs	Yes, at the ensemble level. Significant for both personas; every model changed some answers; no detected difference between prompt set-ups. A specific persona is associated with larger shifts than a generic author line.
RQ3 alike?	Only partly. Models lean the same way as people (weakly), but people shift about three times as much, and human votes spread out while model votes show no detectable spreading.

Hypotheses

	Evidence	Verdict
H1	Author-relevant: significant for A and B. Author-independent: A yes, B no. Unambiguous: nothing detected. (Old method: 0 of 6.)	Partially supported
H2	Both personas $p = 0.0002$; every model changes 18–26% of its answers; a specific persona is associated with shifts about twice as large as a generic author line.	Supported (ensemble level)
H3	Raw: author-relevant shifts 6.2 points more, $p = 0.040$. After removing what chance gives: 1.5 points, $p = 0.70$. Old method: $p = 0.47$.	Weak / not robust
H4 direction	Direction similarity 0.11 vs about 0 by chance, $p = 0.0001$; defined for 2,020 of 3,600 comparisons.	Qualified support (partial, as predicted)
H4 size	Humans 53% vs models 18%; gap 34.9 points, still 33.4 with equal votes.	Qualified support (sizes differ)
Prompt format	No detected difference ($p = 0.199$). No hypothesis in the proposal.	No difference detected (exploratory)

The reasoning behind each verdict

H1, partially supported. H1 makes three predictions. Author-relevant texts shift: yes. Clear texts do not: yes, nothing detected. Author-independent texts shift: only for one of the two personas. Two out of three hold fully, the third halfway.

H2, supported at the ensemble level. The combined test is significant for both personas. Every model changed some of its answers, but those per-model rates are descriptive, so the claim is about the ensemble, not a test of each model separately.

H3, weak / not robust. In the main (raw) test, author-relevant texts shift 6.2 points more ($p = 0.040$). But those texts are the most disputed to begin with, and disputed texts look as if they shift a lot even when nothing real happens. In the post-hoc sensitivity check that allows for this, the gap is 1.5 points ($p = 0.70$). So the conclusion depends on the reference used, and presenting H3 as simply "supported" would be misleading. The gap still points the right way, so it is not disproved, just not established. Remember too that author-relevant texts were defined by a pilot flip test.

H4, qualified support. H4 predicted *partial* alignment in direction and a *difference* in size, and the data show both. It would have failed if the directions were random or if the sizes matched. The qualifications: direction is only defined where both sides moved (2,020 of 3,600 comparisons), and humans and models were measured differently (different people per version versus the same model on both versions).

Caveats to state once

- Each human version was rated by **different people**, while each model answered both versions itself. Part of the human shift is simply different people disagreeing.
- Each model gave **one answer per condition**, so a model's own randomness is not measured.
- About 3 raters per human version is small. Several human–model differences (ties, new emotions) are partly a matter of counts.
- People rated about 30 texts each, but who rated what was not kept, so this link cannot be modelled. It may make the ranges slightly too narrow.
- The extra analyses (chance adjustment for H3, generic-author control, per-emotion tests) were added after the main tests. They are reported next to them, not instead of them.

Say it like this (closing): Author framing changes emotion judgments in both people and language models, most clearly for author-relevant ambiguous texts. Models follow people in direction, but only weakly, and they move much less. The idea that author-relevance specifically drives the effect is suggestive, but not established. Keeping the full vote distribution did two things: it detected human effects that a one-label analysis did not, and it showed that the H3 result is not robust.

Part 7: Likely questions, with short answers

Why not just use the majority label like everyone else?

Because it throws information away. With labels, 0 of 6 human tests are significant; with full distributions, 3 of 6 are. Across all comparisons, 26–41% moved while the label stayed the same; among the comparisons the label method calls unchanged, that is 82% for humans and 51% for models. And for tied texts the label depends on an arbitrary tie rule.

Humans shift 45% even on clear texts. Isn't that huge?

Most of it is ordinary disagreement between different people. Two no-persona measurements of the same texts already differ by about 33%. That is why every test compares against what chance gives, and clear texts show no significant effect.

How can a persona work for A but not B?

Both push in the same direction (+5.4 and +3.0 points above chance). The effect is small and the two results fall either side of the 0.05 cut-off. We did not test A against B, so we cannot claim one persona set worked and the other did not.

So is H3 wrong?

Not wrong, not established. The main test is significant (+6.2 points, $p = 0.040$), but a post-hoc sensitivity check that allows for how disputed each text already is gives +1.5 points, $p = 0.70$. The ordering points the right way, but the result depends on the reference used.

Isn't the human–model gap just because humans have fewer votes?

Unequal vote counts do not plausibly explain it: cutting both sides to 3 votes changes the gap only from 34.9 to 33.4 points. That check does not identify the cause of the gap; the different-people versus same-model design remains a possible contributor.

Are models reacting to the persona or just to seeing any author?

Both, but more to the persona. Neutral to generic author is a shift of about 8%; generic author to specific persona is about 16–19%. These distances do not add up, so they are not shares of one total.

Does the prompt format matter?

No statistically detected difference across the six set-ups ($p = 0.199$). That does not prove there is none, but the effect does not appear to depend on one prompt.

Which model is most sensitive?

Qwen3 changed most often (26.1%), Gemma3 least (18.0%). With one answer per condition these are descriptive rates; no test compared models, so no model is shown to be significantly more sensitive.

Could the same person have rated both versions of a text?

No. No participant saw two versions of the same text. What cannot be recovered is who rated which *different* texts; that may make the ranges slightly optimistic.

How were the three groups decided?

By a fixed rule: at least 4 of 5 original readers agreeing with the author = unambiguous; of the rest, a pilot persona test decided author-relevant (the reading flips) vs. author-independent (it doesn't).